

The Palestinian and Tiberian Vocalisation Traditions of Biblical Hebrew.
A Study of Three Manuscripts with Double Vocalisation from the Cairo Genizah.

Samuel Blapp

Bühlstrasse 4, 8620 Wetzikon

samuel.blapp@students.unibe.ch

Mat. Nr.: 08-066-169

Supervisor: Prof. René Bloch, Faculty of Theology, University of Berne

Co-supervisor: Prof. Geoffrey Khan, Faculty of Asian and Middle Eastern Studies, University
of Cambridge

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Abstract

This master's thesis deals with three biblical manuscripts from the Cairo Genizah (T-S A 19.11; T-S AS 63.6; T-S AS 63.59), that are written with Tiberian and Palestinian vocalisation and accentuation signs on the same words. Comparing them one can still find traces of so-called standard Tiberian tradition, although the usage of Palestinian signs and other Tiberian vocalisation features would point to the non-standard tradition. The placing of the Palestinian accentuation signs seems to show a different understanding of marking stress. Furthermore, vocalisation features, that were generally thought to represent Babylonian influence (i.e. pataḥ/segħol interchange), appear here also in the Palestinian tradition. This thesis also contains a research overview, an introduction to the vocalisation systems of Biblical Hebrew with special regard to the influence of the Syriac vocalisation system, as well as an excursus on shorthand writing.

Corrections

p. 15, note 87

"[...] since this [is] the first time that in Syriac a more elaborate set of different vowel graphemes for [...]."

p. 31, note 163

"The masoretic note to this combination states: יוּ מִיחַד מֶן, which can be translated as "16 unusual occurrences of מֶן."

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1. Introduction

In this thesis I will deal with two vocalisation traditions of medieval Biblical Hebrew. Hence, I intend to examine three manuscripts, that are vocalised not only with the commonly known Tiberian graphemes but also with Palestinian signs. I shall show the differences of the Tiberian vowel notation to Standard Tiberian, which is known from such famous manuscripts like the Leningrad Codex B19a (in the following B19a) or the Aleppo Codex.¹ Moreover, I will compare the usage of the Palestinian signs of these manuscripts with the hitherto published material. It is my intention to contribute to the understanding of the historical development of Hebrew vocalisation systems. The main question that I had in mind is to which extent these double vocalised manuscripts had been influencing the pronunciation tradition of each other. Furthermore, vocalisation is a valuable aid in order to classify manuscripts according to the geographical and chronological context. Although, the two vocalisation systems are not always marked by the same scribe, the reflected pronunciation tradition often reveals the origin and education of its scribe.

I selected for this purpose three manuscripts. Each of them is showing different features. Thus, the first manuscript with the signature T-S AS 63.6 is tiny but giving prove of a scribe, who knew both vocalisation systems. The second manuscript T-S A 19.11 is larger and containing the Palestinian and Tiberian vocalisation and accentuation signs, that are marked by two different *naqdanim*. The third manuscript T-S AS 63.59 is written entirely by one scribe, who uses the Tiberian system with the exception for the /o/ sound, for which he uses the Palestinian grapheme. Furthermore, this manuscript is written in shorthand, which promises some interesting discoveries. Following the examination of this manuscript, I will give an introduction to the history of shorthand-writing of Hebrew. In order to have acquired the required knowledge for the examination of the manuscript, that I made according to a scheme, I will give a short introduction to the Palestinian system and the other vocalisation and accentuation systems of Hebrew. Furthermore, I will summarise the history of its origins and therefore also write about the influence from other languages. At the end I will present

1. Recently, a book has been published about story of the Aleppo Codex and how it was transferred to Israel Friedman, M. 2012. *The Aleppo Codex. A True Story of Obsession, Faith and the Pursuit of an Ancient Bible*. Algonquin Books: Chapel Hill, North Carolina.

my conclusions and suggestion for further research. But before I shall start now, I will give an overview of the hitherto conducted research on the Palestinian vocalisation, after having pointed out some *formalia*.

All references are given in their full extent, when they are quoted for the first time. Thereafter, I shall give "name, year and if necessary page number".² Prof. Khan granted me access to his articles, which will appear in the *Encyclopedia of Hebrew Language and Linguistics*, by sending me its prepublication versions via email. They are referred to with "name, year and title". The comparison of punctuation features with the BHS³ and other primary resources will be indicated in every case separately. Translations are provided by me, if not stated otherwise.

All three manuscripts are held in the Cambridge University Library. Thus, the signature T-S is referring to the Taylor-Schechter collection of documents from the Cairo Genizah, that have been brought to Cambridge in the late 19th century by Dr. Solomon Schechter with the financial aid of Dr. Taylor.⁴

The use of an appropriate font caused some aesthetic problems. Furthermore, some technical difficulties occurred in displaying signs from Syriac and some further accentuation and vocalisation signs.

2. I will use this abbreviated quotation form instead of *ibid.* and *id.* for the sake of clearness and accessibility.

3. Ellinger, K. et al. (eds). 1983. *Biblia Hebraica Stuttgartensia* (4. Auflage). Stuttgart: Deutsche Bibelgesellschaft. (Via Accordance).

4. Recently, two thrilling books have been published on the breathtaking discovery of the Cairo Genizah: Hoffmann, A. et Cole, P. 2011. *Sacred Trash. The Lost and Found World of the Cairo Genizah*. Nextbook/ Schocken: New York. and Glickman, M. 2011. *Sacred Treasure. The Cairo Genizah. The Amazing Discoveries of Forgotten Jewish History in an Egyptian Synagogue Attic*. Jewish Lights: Woodstock, Vermont. The former head of the Genizah Research Unit published also a book on this topic Reif, S. 2000. *A Jewish Archive from Old Cairo. The History of Cambridge University's Genizah Collection*. Routledge: London/ New York.

2. Research Overview

The discovery of the Palestinian vocalisation tradition is closely connected with the discovery of the Cairo Genizah at the end of the 19th century,⁵ since all known Hebrew manuscripts with Palestinian vocalisation stem from this attic in Old-Cairo.⁶ Thus, Neubauer was the first scholar to publish parts of a manuscript which had been written with Palestinian vocalisation.⁷ However, he did not deal with it in this context, since this article is only about shorthand writing (see below). Shortly thereafter, Friedländer announced: "we have here a new, hitherto unknown system of signs for vowels and accents.". Thus, he was the first who published an examination of the vocalisation of these manuscripts.⁸ He continued his research and published another article in which he not only deals with vocalisation but also with the accentuation.⁹

In 1899 Levias published the first comparative study of the Palestinian vocalisation.¹⁰ In this article he compares the vowel signs given by Friedländer¹¹ with those he found in another manuscript and notes the differences e.g. the grapheme of *ševa*.¹² Furthermore, he gives a first thesis for the graphic origins of the vocalisation and diacritical signs. According to Levias the

5. Yahalom, J. 1997. *Palestinian Vocalised Piyyuṭ Manuscripts in the Cambridge Genizah Collections* (Cambridge University Library Genizah Series 7). CUP: Cambridge. 1.

6. Yahalom. 1997. 1. However, they are not only taken care of in the Cambridge University Library, but also in the Bodleian Library in Oxford, where Neubauer served as librarian (?). 2007. "Neubauer, Adolf". M. Berenbaum et F. Skolnik (eds.). *Encyclopaedia Judaica, Vol. 15 (2nd ed.)*. Macmillan Reference: Detroit. 115).

7. Neubauer, A. 1895. "Literary Gleanings. XII. The Hebrew Bible in Shorthand Writing". *Jewish Quarterly Review* 7/2:361-364.

8. Friedländer, M. 1895. "A Third System of Symbols for the Hebrew Vowels and Accents". *Jewish Quarterly Review* 7/3:564-568.

9. Friedländer, M. 1896. "Some Fragments of the Hebrew Bible with Peculiar Abbreviations and Peculiar Signs for Vowels and Accents". *Proceedings of the Society of Biblical Archaeology* 26/3:86-98.

10. Levias, C. 1899. "The Palestinian Vocalisation". *The American Journal of Semitic Languages and Literatures* 15/3:157-164.

11. Levias is not declaring which particular manuscript from Friedländer he is referring to. Moreover, he also just mentions that he received the permission to examine and publish the other new manuscript from a Syrian gentlemen (Levias. 1899. 157).

12. Levias. 1899. 159. As far as I can see this remains the only explicit comparative study of this subject until Revell's publication in the second half of the 20th century (see below).

Palestinian graphemes derived from Nestorian Syriac¹³ and Arabic.¹⁴ Thus, he thinks that the Palestinian *pataḥ* (–)¹⁵ is derived from the Syriac equivalent called *peṭaḥa*¹⁶ and the Arabic *fatha* ˘.¹⁷ Furthermore, he states that the Palestinian *qamaṣ* (˘) sign is also taken from the Arabic *fatha* that represents a small version of 'alif (ا).¹⁸ Moreover, Levias sees this confirmed by the fact that short *qamaṣ* i.e. *qamaṣ qaṭan* (Levias: קמץ חטוף) is represented by ˘ the Palestinian sign also used for [o] sounds i.e. *ḥolam*. He also notes that in this manuscript not only *qamaṣ* and *pataḥ* are interchangeably used but also that there is only one grapheme for *šere* and *seghol*. These phonological peculiarities are known today as typical features of the Palestinian (and Sephardic)¹⁹ vocalisation tradition. Another theory by Levias is that the Palestinian *dages*²⁰ – is derived from the *tašdid*²¹ sign in the Maghribi ductus²² ˘.

According to Yahalom, the majority of the known manuscripts with Palestinian

13. One has to distinguish in Syriac between the Eastern and Western tradition, which does neither have an identical set of consonantal graphemes nor vowels that are marked with the same signs (Brockelmann, C. 1912. *Syrische Grammatik* (Porta Linguarum Orientalium 5). von Reuther & Reichard: Berlin. 5; 9. Segal. J. B. 2004 (Repr. 1953). *The Diacritical Point and the Accents in Syriac* (2nd ed.). Gorgias: Piscataway. 24-36; Muraoka, T. 1997. *Classical Syriac* (Porta Linguarum Orientalium, Neue Serie 19). Harrassowitz: Wiesbaden. 3; 7. However, they have some diacritical signs in common, that already existed before the Schism into Nestorian (Eastern Syriac/ Persia) and Monophysites (Western Syriac/ Byzantine) (Segal. 2004. 22; 24; 37).

14. Levias. 1899. 159.

15. I did not find evidence for this grapheme in the Syriac grammars by Brockelmann and Muraoka. Moreover, according to Friedländer and all the other later scholars (see below) the Palestinian *pataḥ* has always the shape of a horizontal line (–) (Friedländer. 1895. 564; 1896. 98)

16. Muraoka. 1997. 7.

17. Levias. 1899. 159.

18. Cf. The so-called *madda* ˘ in Arabic grammar consists of a horizontal stroke, that represents an 'alif and is used above another 'alif in order to avoid two of them in a row i.e. ٱ for [ā] (Fischer, W. 2006. *Grammatik des Klassischen Arabisch* (4th ed.) (Porta Linguarum Orientalium, Neue Serie 11). Harrassowitz: Wiesbaden. 11).

19. Morag. Sh. 1986. "The Pronunciation of Hebrew in Medieval Spain: Some Notes on Its Early History". D. Muñoz Leon (ed.). *Salvacion en la Palabra. Targum – Derash – Berith* [Festschr. A. Diez-Macho]. Ediciones Cristianidad: Madrid. 749; cf. especially for the pronunciation of *qamaṣ* Blau, J. 2010. *Phonology and Morphology of Biblical Hebrew. An Introduction* (Linguistic Studies in Ancient West Semitic, Volume 2). Eisenbrauns: Winona Lake. 108 (§ 3.5.3).

20. E.g. Dietrich, M. 1968. *Neue palästinisch punktierte Bibelfragmente*. Brill: Leiden. 88*. Revell examines diacritical signs of every manuscript (Revell. E. J. 1970. *Hebrew Texts with Palestinian Vocalization*. University of Toronto Press: Toronto/ Buffalo).

21. Fischer. 2006. 12.

22. Cf. van den Boogert, N. 1989. "Some Notes on Maghribi Script". *Manuscripts of the Middle East* 4:32.

vocalisation has been published by Kahle before the the middle of the 20th century.²³ Thus, he published in 1901 an extensive article on the Palestinian vocalisation in so-called shorthand writing (see below).²⁴ Thereafter, he continued his research on manuscript with this just recently discovered vocalisation tradition and published it in *Masoreten des Westens*.²⁵ The *corpus* does not only include Biblical manuscripts but also *piyyuṭ* (liturgical poetry). Aimo Murtonen continued Kahle's research on the Palestinian vocalisation. His dissertation, which he wrote in collaboration with G. Ormann, contains a Grammar of Hebrew with Palestinian vocalisation.²⁶ Since he is not only using Biblical manuscripts but also *piyyuṭim*, which very likely reflect the pronunciation at this time and give evidence for Medieval Hebrew, this dissertation fills some gaps in Hebrew language studies. However, he arranges his dissertation like a grammar and does not enough take into account, that the vocalisation tradition he is writing about cannot be referred to without showing the differences in vocalisation in each manuscript. I think that this work would have benefitted from Revell's (see below) publications enormously. According to Yahalom, the discovery of the New Series of the Taylor-Schechter Genizah collection revealed further manuscripts with Palestinian vocalisation.²⁷ The Biblical fragments thereof were published by Dietrich in his dissertation.²⁸ Also Dièz-Macho published between 1954 and 1971 regularly on the Palestinian vocalisation partially based on manuscripts from the New Series. These articles primarily deal with the vocalisation, accentuation and other diacritical signs. However, he gives also some details about the content and the manuscript itself.²⁹

23. Yahalom. 1997. 1.

24. Kahle, P. 1901. "Beiträge zur Geschichte der hebräischen Punktation". *Zeitschrift für alttestamentliche Wissenschaft* 21:273-317.

25. Kahle, P. 1967 (Repr. 1927). *Masoreten des Westens* (2 vols.). Olms: Hildesheim.

26. Murtonen, A. 1958. *Materials for a Non-Masoretic Hebrew Grammar I. Liturgical Texts and Psalm Fragments Provided with the So-Called Palestinian Punctuation* (with a contribution by G. J Ormann). Helsinki.

27. Yahalom. 1997. 1.

28. Dietrich. 1968.

29. E.g. Dièz-Macho, A. 1963. "Un nuevo ms. 'Palestinense' del libro de Jueces entre los fragmentos de la Genizah de la Bibliotheca Universitaria de Cambridge". *Seferad* 23/2:236-251 or Dièz-Macho, A. 1953. "Nuevos fragmentos del Targum Palestinense". *Sefarad* 15/1:31-39. For more references see Revell, E. J. 1970. *Hebrew Texts with Palestinian Vocalization*. University of Toronto Press: Toronto/ Buffalo. 202 and Revell, E. J. 1977. *Biblical Texts with Palestinian Pointing and Their Accents*. Scholars Press: Missoula, Montana. 261.

In 1970 E. J. Revell made the major contribution regarding the study of Palestinian vocalisation that still can be considered as such until today.³⁰ He elaborates in this work a classification scheme of twelve categories of the Biblical and non-biblical manuscripts with Palestinian vocalisation based on the usage of the signs. However, this is not the only achievement by Revell in this book, since he also makes the material available, that is needed for the study of this vocalisation tradition. Revell publishes in 1977 another important work on this topic,³¹ but his time he focusses on the accentuation, which he only treats on three pages in his 1970 publication.³² He painstakingly examines all the signs in manuscripts with Palestinian vocalisation, that are used to represent Tiberian accentuation graphemes. Revell develops seven categories for the manuscripts according the form and use of the accents. Furthermore, he gives details about the usage of the accents in comparison to the Tiberian system and notes the differences and developments. One important statement of his conclusion is, that the Palestinian tradition tries "to reestablish a general pattern of non-final stressing that existed in an earlier stage of the language."³³ Besides these two books Revell published also a few articles on the same topics.³⁴ Though, Yahalom states: "Despite these achievements, however, work still remains to be done."³⁵ Yahalom's contribution to this field consists of his catalogue on the *piyyuṭim* with a analysis of the research, which has been conducted so far. He concludes that the usage of vowel graphemes developed from five (i.e. one sign for each sound [a, e, i, o, u]) or six (two signs for *pataḥ* and *qamaṣ*) to seven signs with an additional sign for *šere*. This shows an adaption to the Tiberian system.³⁶ However, the Palestinian systems never developed independent signs neither for *ševa* nor for the

30. Revell. 1970.

31. Revell, E. J. 1977. *Biblical Texts with Palestinian Pointing and Their Accents*. Scholars Press: Missoula, Montana.

32. Revell. 1970. 96-98.

33. Revell. 1977. 214.

34. Including: 1970. "Studies in the Palestinian Vocalization of Hebrew". J. W. Wevers et D. B. Redford (eds.). *Essays on the Ancient Semitic World*. Toronto UP: Toronto. 51-100; and 1972. "The Placing of the Accent Signs in Biblical Texts with Palestinian Pointing". J. W. Wevers et D. B. Redford (eds.). *Studies on the Ancient Palestinian World*. Toronto UP: Toronto. 34-45. For a detailed Bibliography of Revell see <http://english6.net/b/bibliography---articles-and-books-by-e.-j.-revell-w6924.html> [accessed 26.08.13].

35. Yahalom. 1997. 2.

36. Yahalom. 1997. 8.

ḥatefim vowels. Thus, this development is only a gradually adaption of the Tiberian system. Furthermore, Yahalom believes that a classification should be made not only based on the two graphical sets of vowels i.e. six and seven, but also on the usage of the vowel signs i.e. the pronunciation.³⁷

37. Yahalom. 1997. 9.

3. The Historical Background of the Vocalisation Systems of Biblical Hebrew

3.1. Introduction

In the following paragraphs I am going to give a short introduction to the vocalisation systems of Biblical Hebrew and thereby define some important terminological details. However, I will focus on the Palestinian system and use it to approach the other vocalisation systems. A first general observation regarding all vocalisation of Biblical Hebrew has to be made: "It is important to note, however, that none of the medieval traditions appears to be a direct descendant of the earlier attested traditions."³⁸ However, all of them have preserved some features which show traces of earlier pronunciations.

So far, three different systems³⁹ of vocalisation of Biblical Hebrew in the developed Aramaic script (i.e. so-called Jewish script)⁴⁰ have been unearthed.⁴¹ The first evidence, literary speaking, for these three systems might be already given by Maḥzor Vitry (appr. 12th)⁴². 1.) The Babylonian vocalisation is referred to as לִנְקוּד שְׁלֵנוּ, since the origin of this paragraph is Babylon, 2.) the vocalisation signs still used today are mentioned as נִקּוּד טַבְּרִי⁴³, i.e. the same name they are bearing today, 3.) לִנְקוּד אֶרֶץ יִשְׂרָאֵל, which is understood as pointing to the "latest" discovery, the Palestinian vocalisation.⁴⁴ However, historically

38. Khan, G. 2013. "Biblical Hebrew: Pronunciation Traditions". G. Khan (ed.), S. Boložky et al. *Encyclopedia of Hebrew Language and Linguistics*. Brill: Leiden. (forth.).

39. A brief definition of terms: the term "system" refers to a graphic set of vowel signs only whereas the term "tradition" is used to explain how a system is used e.g. As shown by Revell (1970; see below) the Palestinian systems contains of more than one pronunciation tradition as does the Tiberian system (see below).

40. Yardeni, A. 2010. *The Book of the Hebrew Script, History, Palaeography, Script Styles, Calligraphy & Design* (3rd English edition). Carta: Jerusalem. 41-46.

41. The Samaritans, who write also in a Hebrew script until today, that, however, is derived from ancient Hebrew, use also a set of signs for their vowels and diacritics (Kahle, P. 1926. "Die Lesezeichen bei den Samaritanern". C. Adler et A. Ember (eds.). *Oriental Studies. Published in Commemoration of the Fortieth Anniversary (1883-1923) of Paul Haupt*. Johns Hopkins/ Hinrich'sche Buchhandlung: Baltimore/ Leipzig. 428).

42. Goldschmidt, E. D. "Maḥzor Vitry". M. Berenbaum et F. Skolnik (eds.). *Encyclopaedia Judaica, Vol. 13* (2nd ed.). Macmillan Reference: Detroit. 366-367.

43. Actually, it is written according to Luzzato נִקּוּד טַבְּרִי. However, Luzzato thinks it should be emended to נִקּוּד טַבְּרִי (Graetz, H. 1998 (Repr. 1909). *Geschichte der Juden. Von den ältesten Zeiten bis auf die Gegenwart. Band 5: Vom Abschluss des Talmuds (500) bis zum Aufblühen der jüdisch-spanischen Kultur 1027* (4th ed.). Arani: Berlin. 549.).

44. Friedländer. 1896. 94; Kahle. 1901. 275.

speaking, it is assumed the Palestinian and Babylonian systems were invented and used before the Tiberian system was developed.⁴⁵

Table 1.⁴⁶

Standard Tiberian ⁴⁹	Palestinian	Babylonian ⁴⁸	Samaritan ⁴⁷
ⲁ [ɔ]	ⲁ [a] ([ɔ]) ⁵²	ⲁ [a] (ⲁ) ⁵¹	ⲁ [ɔ] ⁵⁰
Ⲃ [a]	Ⲃ [a] ([ɔ]) ⁵³	Ⲃ [ε] (Ⲃ)	Ⲃ [a]
ⲃ [ε]	ⲃ [ε] ([e]; [ʔ]) ⁵⁵	ⲃ [ε] (ⲃ)	ⲃ [e] ⁵⁴

45. Friedländer. 1896. 95.

46. This table is based on Blau, J. 1976. *A Grammar of Biblical Hebrew*. Harrassowitz: Wiesbaden. 9. It consists of the graphemes for the full vowels. Particular signs for ultrashort vowels occur only in Babylonian (see note below) and Tiberian (cf. Dotan. 2007. 624-648 (§5)).

47. It is assumed that the Samaritan vocalisation system first employed also only a diacritical point in order to distinguish homographs, i.e. between vocalic and consonantal value, before it was more elaborated in the Middle Ages most likely after the spread of the Tiberian system (Sáenz-Badillos, A. 2006 (Repr. 1993). *A History of the Hebrew Language. Translated by John Elwolde*. CUP:Cambridge. 152).

48. Two sets of Babylonian vocalisation exist. The more frequent set partially consists of signs that are originally based on parts of letters i.e. 'ayin and 'aleph, and partially of signs represented with dots. The second set, that is attested only rarely, is based entirely on dots and here given in brackets. (Dotan. 2007. 630-631 (§5.2.1.1.1)). Furthermore, under certain circumstances to some of these signs the Babylonian ševa, a horizontal supra-linear line, is added (Dotan. 2007. 631 (§5.2.1.2.1)).

49. Vowel qualities according to Khan, G. 2012. *A Short Introduction to the Tiberian Masoretic Bible and its Reading Tradition*. Gorgias: Piscataway. 82-83.

50. According to Morag the two /a/ sound are sometimes interchanged (Morag, Sh. 1962. *The Vocalisation Systems of Arabic, Hebrew, and Aramaic* (Janua Linguarum 13. Festschr. Nicolai van Wijk). Mouton & Co: 'S-Gravenhage. 42).

51. Cf. Syriac ⲁ [a] that stands only in closed syllables or before a letter with virtual doubling and is attested from the sixth century onwards (Segal. 2004. 28).

52. Through the later influence of Tiberian this sign parallels qamaš. However, the original Palestinian pronunciation tradition does originally not differentiate between the different pronunciations of [a] (Dotan. 2007. 625 (§5.1.1.1) and Yahalom. 1997. 8). This grapheme seems to be introduced later in order to represent Tiberian qamaš (Yahalom. 1997. 8), although it is often interchangeably used with the horizontal stroke of the Palestinian system, that parallels Tiberian pataḥ, this reflects that in Palestinian, which influenced the Sephardic pronunciation tradition, originally only one [a] sound existed (Yahalom. 1997. 8).

53. Dotan. 2007. 625 (§5.1.1.1); cf. also note above.

54. Kahle thinks this sign parallels only Tiberian šere (1926, 428). Furthermore, he suggests that for seghol (and hireq) the same sign was used i.e. that of hireq (Kahle. 1926. 428). Sáenz-Badillos argues that this sign also represented an /a/ (Sáenz-Badillos. 2006. 152).

55. This is the most frequent vowel used to display Tiberian ševa mobile at least in biblical manuscripts (Dotan. 2007. 625 (§5.1.1.1); Dietrich. 1968. 117-121; 88*). However, all vowel graphemes are used to display Tiberian ševa mobile, but only ⲁ, Ⲃ and ⲃ occur regularly (Revell. 1970. 104). Furthermore, the two [e] signs are used interchangeably (Dotan. 2007. 625) cf. also note below.

ֵ [e]	ֶ [e] ([ε]) ⁵⁶	ִ [e] (ִ)	ֶ [e]
ִ [i]	ֶ [i]	ִ [i] (ִ)	ֶ [i]
ֹ; ֺ [o]	ֶ [o]	ִ [o] (ִ)	ֶ [u] ⁵⁷
ֻ; ֽ [u]	ֶ [u]	ִ [u] ⁵⁸ (ִ)	ֶ [u]

3.2. The Historical Development of the Palestinian Vocalisation System

If the Palestinian systems is really that old, how did it develop and what are its origins, since it cannot be derived from the other two systems? Unlike for the Tiberian systems, no grammatical treatise for the Palestinian system is mentioned in any medieval source.⁵⁹ However, like the Tiberian system the other two systems were developed in regions where the Arabic language was used due to the Muslim conquests (around 638 CE the conquest of Jerusalem).⁶⁰ For Tiberian it is said to have been developed only after the Talmud had been completed.⁶¹ The first mentioning of vocalisation is found in the *Diqduqe ha-Te'amim* of Aharon ben Asher, which is said referring to his grandfather, who founded the ben Asher dynasty of Masoretes.⁶² Furthermore, Aharon was not only the *naqdan* of the Aleppo Codex

56. Dotan. 2007. 625 (§5.1.1.1). Originally, like with the [a] sounds only one sign for the two [e] sounds is used (Yahalom. 1997. 8). Though, it is assumed that through the influence of Tiberian Hebrew a further sign was introduced in order to have equivalents that parallel the Tiberian signs (Yahalom. 1997. 8). In Syriac a similar change took place (Segal. 2004. 29).

57. For the Samaritan sounds for [o] and [u] the Arabic sign *ḍamma* ' is sometimes used (Dotan. 2007. 647 (§5.6.1.2); Kahle, P. 1959. *The Cairo Geniza* (2nd ed.). Frederick A. Praeger: New York. 66).

58. Dotan sees this as vertical not diagonal line. Furthermore, this sign can also take the form of a small *waw* (Dotan. 2007. 631 (§5.2.1.1.1)).

59. For medieval grammatical treatises based on the Tiberian system see e.g. Khan, G. 2000. *The Early Karaite Tradition of Hebrew Grammatical Thought. Including a Critical Edition, Translation and Analysis of the Diqduq of ʿAbū Yaʿqūb Yūsuf ibn Nuḥ on the Hagiographa* (T. Muraoka et C.H.M. Versteegh (eds.), Studies in Semitic Languages and Linguistics, Vol. 32). Brill:Leiden/Boston/Köln; and Bacher, W. 1975 (Reprint). *Die Anfänge der hebräischen Grammatik and Die hebräische Sprachwissenschaft vom 10.-16.Jhd. (Amsterdam Studies in the Theory and History of Linguistic Science III. Studies in the History of Linguistics, Volume 4)*. John Benjamins:Amsterdam.

60. Cf. Gil, M. 1992. *A History of Palestine 634-1099 (Translated from Hebrew by Ethel Broido)*. CUP: Cambridge. 51; Safrai, S. et Hirschberg, H. Z. 2007. "Land of Israel (History)". M. Berenbaum et F. Skolnik (eds.). *Encyclopaedia Judaica, vol. 10 (2nd ed.)*. Macmillan Reference: Detroit. 143-179.

61. Arguments for this suggestion are that 1.) vocalisation signs are not directly mentioned in the Talmud nor 2.) that Jerome writes about it (Dotan, A. 1981. "The Relative Chronology of Hebrew Vocalization and Accentuation". *American Academy for Jewish Research Proceedings* 48:89.).

62. Dotan. 1981. 91. Dotan, A. 2007. "Ben-Asher, Moses". M. Berenbaum et F. Skolnik (eds.).

but also mentioned by Maimonides regarding the division of verses in the same copy the complete Hebrew Bible.⁶³ Further evidence sees Dotan given in the colophon of the Cairo Codex (896 CE), in which Moses ben Asher states that the accents derive from a "עדת נביאים" "a congregation of prophets".

According to Dotan this shows that he did not know their origins.⁶⁴ However, evidence from mid ninth century exhibits that parts about it were still known. The Gaon of Shura (853-858 CE), Mar Natronai II Bar Hilai (died 869 CE), states:⁶⁵ לא ניתן ניקוד בסיני כי החכמים ציינוהו לסימן, "the punctuation was not given at Sinai but the sages marked it according to signs". Dotan concludes that already in the early ninth century the precise knowledge about the the origins of vocalisation had been lost and thus, this had to have been invented much earlier i.e. in the seventh century.⁶⁶ Furthermore, he suggests that the accentuation signs were invented much earlier (prior to the fifth cent.), since they are already mentioned in the Babylonian Talmud e.g. Berakhot 62a:⁶⁷ מפני שמראה בה טעמי תורה, "because he shows with it [the hand] the accents of the Torah".⁶⁸ These arguments seem to be convincing, since there is also some evidence from manuscripts, which indicate verse division and other techniques to show the reader how to read it.⁶⁹ However, they are only used as syntactical reading aids and not to indicate a melody. The melody seems to have developed as an mnemonic aid for the

Encyclopaedia Judaica, Vol. 3 (2nd ed.). Macmillan Reference: Detroit. 321. Cf. also Maman, A. 2012. "Ben Asher, Aaron (Abū Saʿīd Hārūn) ben Moses". N. A. Stillman (ed.). *Encyclopedia of Jews in the Islamic World*. Brill Online.

63. ספר שסמכנו בדברים אלו [...] שהיה בירושלים מכמה שנים להגיה ממנו הספרים [...] לפי שהגיהו בן אשר, [...] ועליו [...] "a book, which we support regarding this issue [verse division] [...]. It has been used in Jerusalem for many years in order to correct [other] books [...], since ben Asher revised it [...] and I rely on it for *Sefer Torah*, which I wrote according to his law." (*Mischneh Torah, Hilchot Sefer Torah* 8.4 [Hebrew text] quoted from Maimonides. *Mishneh Torah. Hilchot Tefilin UMezuzah V'Sefer Torah and Hilchot Tzitzit. A new translation with commentaries and notes*. Eliyahu Touger (ed.), Moznaim Publishing Corporation: New York/Jerusalem, 1990. 156.

64. Dotan. 1981. 92.

65. Dotan. 1981. 92.

66. Dotan. 1981. 92.

67. Dotan. 1981. 95-99.

68. Cf. Levin, S. 1968. "The Traditional Chironomy of the Hebrew Scriptures". *Journal of Biblical Literature* 81/1:59-70.

69. Revell, E. J. 1976. "Biblical Punctuation and Chant in the Second Temple Period". *Journal for the Study of Judaism* 7/2:181.

syntactical correct reading of the Bible, which, however, were indicated by signs much later.⁷⁰ Early traces of marking accentuation can be found in Greek manuscripts from Qumran that employ spaces and capital letters where the Tiberian Hebrew system has major disjunctive accents (i.e. *silluq*, *atnah*, *zaqef* and *ṭifḥa*).⁷¹ Thus, the idea of accentuation can be seen as one of the oldest tools to secure a certain interpretation, since they mark syntactical entities.

In my opinion the seventh century as the inaugural date for the notation of qualitative distinctive vowel graphemes seems to be very early.⁷² At that time Syriac just started to develop its different vowel graphemes.⁷³ Furthermore, I do not agree with Dotan's argumentation that supra-linear vocalisation points towards a historically early invention. This is not only confirmed by Hebrew, where diacritical marks are placed above and below a letter,⁷⁴ but also by Syriac where the early diacritical marks (sixth century)⁷⁵ and later vowel signs (mid seventh cent.) have been marked not only supra-linear but also sub-linear.⁷⁶ Unlike, in Western Syriac, where all vowel signs were used in both positions (after the seventh cent.), in Eastern Syriac they had to be marked in a fixed place i.e. either supra- or sub-linear, since their position defined their pronunciation due to the use of graphical identical signs e.g. the single dot could be used to denote either an [e or i] or [a] sound depending on its position.⁷⁷

Neighbouring cultures, like Syriac, started to specify their diacritical marks around 600 CE.⁷⁸ By mid seventh century the East Syrian vocalisation system consisted already of vowel

70. Revell. 1976. 181.

71. Revell, E. J. 1971. "The Oldest Evidence for the Hebrew Accent System". *Bulletin of the John Rylands Library* 54:215.

72. Dotan. 1981. 92.

73. Morag. 1962. 46 (§3.2.1); Segal. 2004. 24-32 (especially 28-29).

74. Morag. 1962. 17.

75. Segal. 2004. 5. E.g. The signs – [e] and ÷ [a] are found in both East and West Syrian manuscripts towards the end of the sixth century (Segal. 2004. 23).

76. Segal. 2004. 24-32.

77. Segal. 2004. 24-32; 44.

78. Segal. 2004. 5. Hebrew like Syriac was using in early stages diacritical marks to distinguish between homographs. However, they are delivered in Hebrew only in Masoretic lists, whereas Syriac employs them regularly in manuscripts (Morag. 1962. 17). Furthermore, in Syriac diacritical marks are attested at least since the fourth century (Morag. 1962. 46 (§3.2.1)).

graphemes that express different vowel qualities.⁷⁹ The West Syrian system by then consisted still of a kind of developed diacritical marks.⁸⁰ Jacob of Edessa developed his own vowel system which was written between the consonants like Greek or other Indo-European scripts.⁸¹ However, this system was not accepted since the texts, that were transmitted had to be changed in order to use this new system. Therefore, the so-called Greek vowel set has been invented, which could be written below or above the consonants.⁸²

Comparing the first Syriac vowel systems (appr. seventh cent.) and the early Palestinian system, one can find some similarities. Graphically speaking, some of the Eastern Syriac vowel signs consist also of a certain combination of dots e.g. $\dot{\text{—}}$ or $\dot{\text{—}}$ or $\dot{\text{—}}$ for the [ɔ] sound, that were introduced around the 8th century.⁸³ In the Palestinian system the vertical dots are connected to a vertical stroke and correspond to Tiberian *qamaṣ* [ɔ] $\dot{\text{—}}$. The other two signs $\dot{\text{—}}$ and $\dot{\text{—}}$ are supra-linearly used in the Palestinian pronunciation tradition to represent [e] sounds (see below).⁸⁴ The sub-linear version of these two signs, i.e. — and — (together with —) were introduced to represent a further vowel sound i.e. [ê] between [ě] and [i], in Eastern Syriac.⁸⁵ The same can be applied to the Palestinian system, where a later introduced [e] sound, that is often used to represent *šere*, is represented by $\dot{\text{—}}$. Yahalom thinks this indicates an influence of the Tiberian system. Furthermore, Segal sees the introduced [e] sign in Syriac

79. Segal, 2004. 29.

80. Segal. 2004. 38-39. Jacob of Edessa writes in his grammar, that are some rules regarding the application of these diacritical signs: "(Vowel) sounds are thick and thin. Again, every word, that is, every member of a clause – where it is thick or broad in (vowel) sound, there it takes a point above (the line); where it is fine or thin (it takes a point) below. If it is medium, (between) fine and thick, and there are two other words similar to it in spelling, it takes two points, one above and one below [...]." (Segal. 2004. 38-39).

81. Segal. 2004. 42-43.

82. Segal. 2004. 44-45.

83. This letter was called in Syriac *z'qāfā*. In biblical Hebrew the same sign, two horizontal dots, were also called *Zaqef*, which is base on the root קָם "to raise, put up, erect" (Jastrow, M. 2004 (Repr.). *Dictionary of the Targumim, Talmud Bavli, Talmud Yerushalmi and Midrashic Literature*. Judaica Treasury:New York. 410).

84. Yahalom. 1997. 8.

85. Yahalom. 1997. 8; Segal. 2004. 29. Kahle also thinks, that they are taken from Syriac (Kahle. 1901. 303). I am not sure to which extent the qualitative and quantitative pronunciations of these different [e] sounds in Syriac are corresponding to those in the (Palestinian) Hebrew pronunciation. Furthermore, the pronunciations in both languages might have changed through the centuries.

as corresponding to Tiberian *šere* or *hireq*.⁸⁶ Moreover, the later introduced [e] sound in the Palestinian system also corresponds with Tiberian *šere*. The Palestinian system uses the two vertical supra-linear dots to represent an [i] sound ֿ. Phonologically speaking, it is difficult to tell to which extent the Syriac vowels correspond to the Palestinian vowels.⁸⁷ *Ševa* is in Western Syriac, like in the Palestinian system, often represented by the same sign which usually denotes an [e] sound.⁸⁸ Though, there are no graphical similarities between Western Syriac and Palestinian Hebrew.⁸⁹

These similarities of Eastern Syriac and the Palestinian system cannot give prove of their connection. It seems as if the the corresponding graphemes in Eastern Syriac and the Palestinian system were used on the other side i.e. supra- and sub-linear, where the corresponding sign was used in the other language e.g. in Syriac ܐ vs. ֿ in the Palestinian system. However, these remain just observations until further research will be conducted. For the Babylonian system no influence by the Syriac vocalisation system can be proven with security.⁹⁰

The Samaritan system of vocalisation⁹¹ contains also some graphical similarities with the

86. Segal. 2004. 31.

87. If one compares the developments of the Syriac vocalisation with the Tiberian Hebrew vocalisation, it could be inferred that, Tiberian Hebrew was influenced by Syriac, or vice versa, regarding the vowel qualities. However, there is no evidence based on manuscripts, that would give proof of vocalised Tiberian Hebrew manuscripts from the sixth or seventh century. Thus, I think it is more likely, that Tiberian Hebrew could not have been invented before mid seventh century or early eight century, since this the first time that in Syriac a more elaborate set of different vowel graphemes for [a] and [ɔ], and [e] and [ě] occurs. However, this is only valid for Eastern Syriac, since the latter phonological distinction is not attested at this time in Western Syriac (Segal. 2004. 44). It is astonishing that the same grapheme in Tiberian and Eastern Syriac is used to represent the [i] sound i.e. ֿ or with *mater lectionis yodh* ֿ. In Syriac the sign that corresponded with *šere* was invented later. However, the same sign for *šere* is used in Tiberian (sub-linear) and in Babylonian (supralinear). The same sign was also used in Eastern Syriac but only until the end of the seventh century, when another [e] sound grapheme was invented to distinguish different sounds. Maybe this shows that the sign ֿ has been borrowed by the Tiberian and Babylonian vocalisation systems by the time it was the only representation of an [e] sound i.e. in the mid seventh century (Segal. 2004. 29).

88. Segal. 2004. 44-45. The position and graphical representation of *ševa* in the Tiberian system corresponds to the East Syriac vowel grapheme for an [e] sound introduced in the eight century, which corresponds to Tiberian *šere* and *hireq* (Segal. 2004. 30-31).

89. On the similarities between the Babylonian Hebrew system and Western Syriac see Dotan, A. 2007. 631 (§5.2.1.1.2).

90. Dotan. 2007. 631 (§5.2.1.1.2).

91. It is noteworthy that the Samaritan grapheme for *seghol* consists of the same shape as the Tiberian counterpart. Unlike, in the latter the former uses lines to form the triangle and not dots i.e. ֿ vs. ֿ. Furthermore, the same can be applied for the Palestinian and Samaritan [o] sound. The former is represented by three dots,

Palestinian system.⁹² 1.) the sign for the [a] sound ֿ, corresponding with the same Palestinian grapheme for the supposedly same sign, 2.) the sign for the [ɔ] sound, which corresponds to Palestinian ֿ. 3.) the usage of ֿ for *dageš* in both systems. Dotan confirms the closeness of some of the vowel graphemes, but sees no necessary connection. Furthermore, he adds that the two accentuation systems differ completely and classifies it closer to the Syriac system.⁹³ Morag sees in the interchange of the Samaritan equivalent of *qamaš* and *pataḥ* not the only resembles of the Palestinian vocalisation tradition.⁹⁴

Although the Babylonian system uses similar vowel signs, they are not directly corresponding in any relation to the Palestinian pronunciation tradition e.g. ֿ is used in the Babylonian system to mark an [o] sound whereas it is used in Palestinian Hebrew for the [i] sound. It seems to me as if they are derived from a common system i.e. Syriac, although Dotan questions such a hypothesis.⁹⁵ However, one can find evidence for corresponding signs of both systems in Western and Eastern Syriac.⁹⁶ Though, these are just observations, which cannot be proven at the moment.

Apart from Syriac, Arabic was the dominant language spoken in this region at the time in question. The influence of Arabic grammarians on Hebrew grammarians has been shown and is therefore evident.⁹⁷ However, Arabic grammatical thought is only traceable from the middle of the eight century CE onwards.⁹⁸ Brockelmann locates its initiation in *Bašra* due to

whereas the latter is represented by lines (Dotan. 2007. 647; Kahle, 1926. 428).

92. Dotan. 2007. 647 (§ 5.6.1.2).

93. Dotan. 2007. 647 (§ 5.6.1.2). For more details on the Samaritan diacritical signs see Kahle 1926.

94. Morag. 1962. 42.

95. Dotan. 2007. 631 (§5.2.1.1.2).

96. Segal. 2004. 29-30; 45.

97. E.g. Khan, G. 2000. *Early Karaite Grammatical Texts* (SBL Masoretic Studies 9). Society of Biblical Literature: Atlanta. 18 (note 9): "Saadya [Gaon] follows the teachings of the *Bašran* school of Arabic grammar that had become mainstream by the tenth century and maintains that the base of derivation is an infinitive or nominal form. [...] According to 'Abū al-Faraj Hārūn, the view that the imperative was the base of derivation was held also by Arabic grammarians of the *Kūfan* school." Furthermore, in the *Sefer Diquḏe ha-Te'amim* of Aharon ben Ašer the Arabic root ʃ-ر-ف (*šarf*) is used to denote the "inflexion" of nouns. (Bacher. 1975. 35; Cowan, J. M. 1952 (Repr. 2011). *Arabic-English Dictionary. The Hans Wehr Dictionary of Modern Written Arabic*. Snowballpublishing. 513).

98. Brockelmann, C. 1943. *Geschichte der arabischen Litteratur (Band 1)*. Brill: Leiden. 96.

variety of languages and dialects spoken there.⁹⁹ Thus, not only Muslims lived there, who were speaking different dialects of Arabic, but also Persian and Aramaic speaking people resided there.¹⁰⁰ According to Brockelmann, two main demographic facts demanded linguistic treatises: 1.) A variety of different dialects coming together and causing communication problems, 2.) the need of non-native speakers to learn Arabic for religious (conversion) or administrative purposes.¹⁰¹

As already mentioned above, Levis believes that also Arabic has influenced the notations of the [a] sounds and that of *dageš* in the Palestinian Hebrew pronunciation.¹⁰² Although, this cannot be proven, it seems to be a valuable hypothesis. This means that the Palestinian vowel notation cannot be dated earlier than the eighth century, as, according to Brockelmann, the first Arabic grammatical works had been written at this time.¹⁰³

Since evidence of Palestinian vocalisation is usually rare it is a painstakingly work to establish theses about it. However, this has been done in the most elaborate way by Dietrich and especially Revell.¹⁰⁴ They managed to study a fair amount of manuscripts that allowed them to draw further conclusions. Although, Dietrich's approach is not as systematic and scientific as Revell's is, he still notes, in my opinion, the important observation, that the usage of the Palestinian graphemes is not coherent. Revell's detailed study of biblical and non-biblical manuscripts with Palestinian vocalisation confirms this observation. He elaborates twelve classes which are arranged according to their usage of the graphemes. Thus, he is able to show that various dialects of Hebrew had been used with Palestinian signs.¹⁰⁵ Moreover, he defines two main groups of dialects. Although, there is no evidence that would give hints as to which community spoke these dialects, Revell suggests that the group A represents the Jews of Palestine, whereas group B represents the Jews of Egypt.¹⁰⁶ Furthermore, he also

99. Brockelmann, 1943, 96f.

100. Brockelmann, 1943, 96.

101. Brockelmann, 1943, 96.

102. Levis. 1899. 159.

103. Of course it is possible that earlier works exist, but they are not known to us.

104. Dietrich. 1968; Revell. 1970; 1977.

105. Dietrich. 1968. 88*; Revell. 1970. 99-121.

106. Revell. 1970. 120. It is assumed that Palestinian pronunciation tradition of Hebrew spread to North Africa and Europe during the middle ages. Hence, it is also known as Sephardic pronunciation tradition.

detects subgroups within these two dialects (i.e. variants in pronunciation), but he does neither consider them as standing for own dialects nor representing chronological stages. It is a combination of features that define group A.¹⁰⁷ For example, the usage of the Palestinian sign, that is usually used to represent Tiberian *qamaš*, which points towards the realisation of [a] sounds as back [a] i.e. [ɔ]. Furthermore, the pronunciation of *ševa mobile* before *yod* as [i]. Both of them are typical characteristics of the ben Asher tradition that has come down to us in the *Sefer Diqduqe ha-Te'amim*.¹⁰⁸ Furthermore, the representation of short *qamaš* by the Palestinian [o] sound can be seen as a genuine feature of the Palestinian system.¹⁰⁹ According to the ben Asher tradition no distinction is made in the pronunciation of the three *qamaš* versions (i.e. *qamaš gadol*, *qaṭan*, *ḥaṭaf*).¹¹⁰ Characteristics of the second group are the different representation of short *qamaš* by [o, a, ɔ] signs and the frequent interchange of *qamaš* and *pataḥ* and *seghol* and *šere*, which is only scarcely attested group A. Both of these phenomena of group B are considered to be influences from other languages, that in Egypt are more likely to happen than in Palestine.¹¹¹

If one compares the different graphemes used for vocalisation and accentuation in the various systems of Hebrew vocalisation, one finds similarities not only regarding the graphic signs but also regarding the pronunciation of vocalisation signs (see Table 1). Thus, for example, in Tiberian, Palestinian and Samaritan a horizontal stroke is used either supra- or sub-linear to denote a simple [a].¹¹² The horizontal line is in two systems used to represent a back [a] sound (i.e. Palestinian and Samaritan). Furthermore, two horizontal dots occur three times in three different vocalisation systems to denote two different vowel qualities (i.e [e] in Tiberian (sub-linear), Babylonian (supra-linear) and [u] in Palestinian). The Babylonian sign for [o] is used in the Palestinian system for an [i] sound and the inverted Tiberian *seghol* is

Moreover, it spread during the same time also in the other direction towards the East (i.e. today Iraq and Iran) and replaced there the Babylonian system (Khan. 2013. Biblical Hebrew: Pronunciation Traditions).

107. Revell. 1970. 113.

108. For §12 on *ševa mobile* before *yod* see Bacher. 1975. 28-29.

109. Morag. 1986. 749.

110. Blau. 2010. 109 (§3.5.3.6).

111. Revell. 1970. 113-115.

112. In the Babylonian systems the same sign is used to denote *ševa mobile*, which was also pronounced as [a] sound (Dotan. 2007. 631 (§5.2.1.1.1)).

used in Palestinian Hebrew to represent an [o] sound. In Samaritan these three dots are connected in order to form a "roof" i.e. ^, which represents an [u] and [o] sound. The two Palestinian signs for the [e] sounds are also attested in the Babylonian vocalisation system. Though, only the Palestinian sign $\dot{\text{e}}$ [ɛ] could also be used in the Babylonian system to denote [ɛ]. Moreover, both of them are used for Tiberian *seghol*.¹¹³ The other grapheme $\dot{\text{u}}$ is used in the Babylonian system to represent [u], whereas it stands in Palestinian for [e]. However, such observations can also be made within a system e.g. in Tiberian where the inverted *seghol* is used to denote the postpositive disjunctive accent *segholta*. However, in the Tiberian system identical signs are not used for different purposes in the same positions,¹¹⁴ whereas in the Palestinian system several signs are used, that could either represent a vowel or an accent e.g. *pašta* which is represented by the same sign as the [u] sign.

Thus, due to the identical signs that are used one might think of an evident connection between the different systems. However, such a connection is very difficult to establish and has to be based on a detailed study of each grapheme i.e. the full background of the usage of a vowel has to be elaborated in order to start to trace a possible influence from the one to the other system. The same applies to Syriac and the similarities that can be found in the vocalisation systems of Hebrew.

The Palestinian system consists only of full vowel signs. Thus, they indicate only vowel quality and not quantity.¹¹⁵ Furthermore, the usage of these graphemes has never become standardised like the Tiberian system. Hence, almost every manuscript shows some individual characteristics regarding the usage of the vowel signs. This is also confirmed by the usage of the full vowel signs to denote *ševa mobile* (see Table 1). Generally speaking, between five and seven vowels can be employed in this system. Yahalom thinks (see above) that the Palestinian system started with five and developed through a process of adaption of the Tiberian pronunciation system to seven¹¹⁶ graphemes. However, he does not indicate the

113. See above Table 1: Babylonian knows only one sign for Tiberian *pataḥ* and *seghol*.

114. *Mappiq* (hebr. for "pronunciation") is used with *he* (and sometimes *'aleph*) in Tiberian Hebrew like *dageš lene* with the BGDKFT letters (Dotan. 2007. 635 (§5.3.2.1.2)).

115. Dotan. 2007. 625 (§5.1.1.1).

116. Sometimes even eight graphemes are attested e.g. Dietrich shows the usage of an additional sign for *ševa mobile* in a biblical manuscript. However, this is borrowed from the *dageš forte*, which is denoted with the same

resources for this information. Unlike Dotan I do not agree with him that the Palestinian system is known to us only in a developing stage,¹¹⁷ at least not in the meaning of "developing" as he understands it. As far as I understand his thesis, he uses the term "development" as a point in the history of the development of a system towards standardisation. However, I think one can find evidence for a development regarding the number of signs, that are used, but there is no development towards a standardisation. Even for the representation of the Tiberian system no distinction of quantity ever seems to be regarded as necessary. Thus, it never became standardised, since it was not intended to or no need has been felt to standardise the pronunciation, but rather to serve as a reading aid. Revell also finds no evidence to conclude that the different ways of vocalising show historical stages. He thinks that the differences can be explained by linguistic developments within the two dialects of Palestinian Hebrew.¹¹⁸ Furthermore, Dotan argues that early manuscripts contain less signs, relating to the frequency of marking vowels and not to the number of phonetic distinctive graphemes.¹¹⁹

My knowledge about classifying manuscripts according to the script is limited, but I think that T-S A 19.11 could be seen as an example for Dotan's observation. Comparing the shape of ה in this manuscript with Yardeni's examples for the 9th century one could come to this conclusion.¹²⁰ I think this is the oldest of the three manuscripts (9th century). Furthermore, I would conclude that the T-S AS 63.59 is the youngest (about 10th century) and T-S AS 63.6 was written in between these two. However, it is very difficult to give such information, since it is based on the shape of letters, that were constantly developing and show even within one type of handwriting differences. For the latter two I also compared the letter ה, which seems to show a development within these three manuscripts as shown by Yardeni.¹²¹ Furthermore, also the usage of the Tiberian vowel signs points towards a later date of T-S AS 63.59 (see below).

sign in this manuscript (1968, 88*).

117. Dotan. 2007. 625 (§5.1.0.2.).

118. Revell. 1970. 120.

119. Dotan. 2007. 625 (§5.1.0.2.).

120. Yardeni. 2010. 209-219.

121. Yardeni. 2010. 209-219.

3.3. The Tiberian Vocalisation System

Since I will examine manuscripts with both, the Palestinian and Tiberian system, I would like to give a brief introduction to the latter here. Usually, the Leningrad Codex B19a and the Aleppo Codex are mentioned first if it comes to the question to name manuscripts with Tiberian vocalisation. However, there are several others less precious manuscripts with Tiberian vocalisation, that are sometimes consisting only of a few lines. Most of these manuscripts are available only since the discovery of the Cairo Genizah. It took some time until scholars discovered that, although written with Tiberian signs, some of these manuscripts used these signs in different manners than B19a that became famous due the BHS (BHK), which entirely relied on it.¹²² Thus, two explanations are given 1.) they reflect a different pronunciation tradition than the one found in e.g. B19a or 2.) they prove a different method of notation and different rules for the usage of Tiberian signs.¹²³ Since the vocalisation of B19a and the Aleppo Codex was considered as representing "the most perfect and elaborated" system of vocalisation, and thus was thought to represent a standardised stage, scholars refer to it, when they are writing about Tiberian vocalisation or the Masoretic text. The colophons of these two manuscripts and the findings of Masoretic works stemming from the vocalisers of B19a and the Aleppo Codex (i.e. the ben Asher family, see above) proved the existence of a school and thus confirmed the standardised vocalisation of these two versions of the Hebrew Bible. Though, they are not entirely standardised i.e. for example coherent in marking *rafe* and secondary stress (*ga'ya*).¹²⁴

Various names have been suggested for this other newly discovered system. All of them are based on features of vocalisation and accentuation of such manuscripts.¹²⁵ Kahle, for

122. The new edition BHK still relies on B19a, but gives variants in reading from other important manuscripts.

123. Dotan. 2007. 642 (§5.4).

124. Dotan. 2007. 635 (§5.3.2.2.2).

125. Cf. Dotan. 2007. 642-644 (§5.4.1). Some of these features, which Dotan mentions are 1.) *dageš lene* is used not only on the BGDKFT letters but also in every letter of a word, that is preceded by a word with a closed syllable, 2.) *pataḥ furtivum* is often represented only by the sign of *ševa* (this does not necessarily have to be an example of Non-Standard pronunciation, since *ševa mobile* has been also realised as a short [a] sound in Standard Tiberian). 3.) Rafe is not exclusively used on the BGDKFT letters, but also on other letters e.g. ם in order to mark its consonantal pronunciation. The Codex Reuchlinianus is probably the most famous manuscript to represent such features (see below).

instances, named it "Ben-Naftali"¹²⁶ due to his focus on the change of e.g. הָ to הֵ .¹²⁷ Díez Macho and Sperber considered it more primitive and thus called it "proto-" or "pre-Masoretic". The opposite believe is attested in the term "post-Masoretic". Morag preferred a term including the reference to the Palestinian system due to the interchange of qamaṣ/pataḥ and *seghol/šere*.¹²⁸ Yeivin, suggested at first the terms "non-masoretic", since he considered it not being based on the accepted Masoretic tradition (i.e. ben Asher). The second term he suggested was "Expanded Tiberian", which mainly underlines the Tiberian nature of such manuscripts.

Díez Macho undertook a first attempt to classify manuscripts with the "ben Naftali" vocalisation into three classes, since he thought to have found a development, that could be shown with his classification.¹²⁹ Thereafter, Davis offered a classification scheme for the Tiberian system in general.¹³⁰ He distinguishes between three groups 1.) Standard Tiberian (so-called ben Asher type), 2.) Predominantly Standard Tiberian and 3.) Non-Standard Tiberian.¹³¹ Davis applies the quantitative differences in the usage of the vocalisation signs in comparison to "Standard Tiberian" as main classification feature. However, this classification scheme is only applicable on manuscripts that actually can be related to the Standard Tiberian tradition.

Dotan calls it the "Tiberian Non-Conventional" system, since he finds all other attempts not appropriate. I do agree with him on that, since his term is more precise in being less specific. The previous terminology can only be employed for manuscripts that explicitly show, for example, Palestinian influence. Furthermore, terms like this neglect influence from other systems like the Babylonian (see below T-S A 19.11), if they serve as the only opposing class for manuscripts that are not employing Tiberian graphemes according to B19a or the

126. Kahle. 1967. 45*-68*.

127. Dotan. 2007. 642 (§5.4.1.1).

128. Dotan. 2007. 644 (§5.4.3.2).

129. Díez Macho, A. 1963. "A New List of So-called 'Ben Naftali' Manuscripts". D. Winton Thomas et W. D. McHardy (eds.). *Hebrew and Semitic Studies. Presented to Godfrey Rolles Driver in celebration of his Seventieth Birthday*. Clarendon:Oxford. 16-52.

130. Davis, M. 1978-1980. *Hebrew Bible Manuscripts in the Cambridge Genizah Collections, vols. 1-2*. CUP:Cambridge.

131. Davis. 1978. xii.

Aleppo Codex. In my opinion, "Tiberian Non-Conventional" is a well chosen term to oppose "Conventional Tiberian" for B19a, the Aleppo Codex and similar manuscripts.

However, I think "Tiberian Non-Conventional" system is still to less precise, since manuscripts with various backgrounds are using the Tiberian graphemes i.e. there is not only one additional system but there are several. Thus, I would elaborate a more precise terminology and classification. I could imagine to classify manuscripts according to their pronunciation tradition only. Thus, my classification scheme would consist of several groups. One of them might be called "Palestinian-Tiberian" system, since there are certainly a lot of manuscripts that clearly show such influence regarding the pronunciation. Furthermore, I would also add other groups, for example, a "Babylonian-Tiberian" or "Babylonian-Palestinian" group (see T-S A 19.11). Moreover, I am thinking of specifying the "Conventional Tiberian" group, which could be also called "Standard Tiberian", with subgroups according to their school i.e. ben Asher and ben Naftali.

Furthermore, such a classification should also be based on the frequency of the influence from other systems or the development of a group. T-S AS 63.59, for instance, employs on the one hand the Palestinian sign for [o] although using entirely Tiberian graphemes and on the other hand shows a development of the Standard Tiberian tradition regarding the uses of *ga'ya*, *dageš* and *rafe*. Thus, some manuscripts may only show scarce influence whereas another manuscript might be classified within the same subgroup but using only the pronunciation tradition of this subgroup. Hence, these subgroups should be divided into further subgroups. These are just suggestions based on my research for this thesis. A sophisticated and elaborated classification scheme that is based on evidence from more manuscripts, will be subject of my PhD. Since this was not the purpose of this paper, I am using Davis' terminology and if necessary I will give additional information about certain features.

4. Manuscript Examination

4.1. Introduction

In the following I will present a detailed description of the three manuscripts, which I have examined. This examination will always follow the same pattern (see below). In large parts I will be comparing features of vocalisation that I found in the manuscript with those from the Standard Tiberian tradition as shown by the BHS and the Leningrad Codex B19a (in the following B19a), the manuscript it is based on. Since the BHS is representing B19a only to a certain extent due to technical problems, I will indicate which of the two sources I am quoting from. For more details see my notes in the appendix. It is the basis for my elaborations and in which more examples and further observations can be found. These notes are given for each manuscript and arranged either according to topics or verses. As a rule, I indicated only the relevant vocalisation signs. However, this is not only employed to make it easier and quicker accessible for the reader but also helped me to take my notes faster. As a basic source for the manuscript I used the catalogue from Davis/Outhwaite, in which all my manuscripts are briefly described.¹³²

4.2. Method

In order to display the different features of the manuscripts, they are examined according to the following scheme. Its different parts are intended to show all aspects of manuscript studies, which I tried to deal with to the best of my knowledge. Furthermore, my scheme is inspired by previous studies from Revell (1970) and Dietrich (1968). Revell's publication takes an important part in this examination and will help me to show the manuscripts in a wider context.

4.2.1 Examination Scheme

⌘) **Content.** I shall give here the exact content of this manuscript i.e. book, chapter and verse.

132. Davis, M. et Outhwaite B. 2003. *Hebrew Bible manuscripts in the Cambridge Genizah collections*, vols. 3-4. CUP:Cambridge.

ב) **Language; vocalisation; accentuation; ink.** In order to give a short introduction to every manuscript, I will show with which system of vocalisation and accentuation one has to deal in this manuscript. Descriptions of the colour of the ink can be very useful in order to differentiate between two different hands of which e.g. one wrote the Palestinian and the other the Tiberian signs.

ג) **Material; size; leafs; columns; lines; physical description of the manuscript; script.** These features are necessary when dealing with manuscripts, since they can give important information regarding the origin of a manuscript. However, I did not focus on this part and therefore these information are given but not extensively elaborated.

ד) **Words with double vocalisation.** I will list all words that occur in a manuscript in order to make it easier to compare the vocalisations.

ה) **Palestinian vocalisation graphemes used in this manuscript.** All the palestinian vocalisation signs will be listed separately, supported by examples and compared with their counterparts of the Standard Tiberian vocalisation tradition.

ו) **Palestinian disjunctive accents.** As with the vocalisation signs I will also give detailed evidence for the occurring disjunctive accents and compare them to the usage in Standard Tiberian. However, such a comparison is always based on the assumption that the same division of the accents can be applied in both systems. Furthermore, the disjunctive accents are arranged according to Yeivin's classification of four grades,¹³³ which is not depending on their pausal value but rather on the pattern they are used in sentences i.e. "a unit ending with a disjunctive of grade I is divided by one of grade II, not by one of grade III, and so on."¹³⁴

ז) **Palestinian conjunctive accents.** The same method as described in ה and ו will be applied but with the disjunctive accents.

ח) **Palestinian diacritical signs used.** Here, I shall list the Palestinian signs that are reflecting Tiberian signs like *dageš*, *rafe* and other signs that do not fit into the categories ה to ז.

ט) **The Palestinian vowels according to Revell's classification.** As already mentioned

133. Yeivin, I. 1980. *Introduction to the Tiberian Masorah (translated by E. J. Revell)*. Scholars Press. 169 (§197).

134. Yeivin. 1980. 169 (§197).

above, I will try to classify the vocalisation tradition of a manuscript according to Revell's classes, which he has elaborated for the Palestinian vocalisation tradition.¹³⁵

י) **The Tiberian vocalisation tradition.** In this section I will give a detailed analyses of the vocalisation represented by Tiberian vowel signs and elaborate what kind of Tiberian pronunciation tradition is reflected.

יא) **The Palestinian accentuation systems and its classification according to Revell.** Like in the part on Palestinian vocalisation, I will try to classify a manuscript based on the observations given in י and יא according to Revell's groups of accentuation.¹³⁶

יב) **Other noteworthy features.** Here, I will list and explain features that do not fit into one of the preceding classes e.g. plene/defective writings, Masoretic notes, shape of letters, verse division, personal comments and other noteworthy observations.

יג) **Summary.** In a short summary of every manuscript I will try to give an overview over the discoveries.

4.3. T-S AS 63.6

א) Judges ¹³⁷6.19-7.9 and 10.12-20.

ב) Hebrew with Palestinian vocalisation and accentuation. Tiberian vocalisation on one word. Consonantal text, vocalisation and accentuation are written in the same ink.

ג) Parchment; 6,4 x 9,7 cm; 1 leaf; 2 columns; 8 lines; Badly mutilated and rubbed (only F); Only a few words legible;¹³⁸ Oriental square script, most likely South-Western type (Egypt or Palestine), e.g. left downstroke of *Aleph* and angular bases of *Beth* and *Pe*.¹³⁹

ד) וַיִּצָא and וַיִּשָּׂא.

135. Revell. 1970.

136. Revell. 1977.

137. Davis/Outhwaite mention that it begins with v. 7.2 (2003, 412 (vol. 4)).

138. Davis/Outhwaite. 2003. 412 (vol. 4).

139. Yardeni, 216-219. Olszowy-Schlanger, J. "Oriental Script". Powerpoint presentations in the course of the workshop "Hebrew Manuscript Studies: An Introduction". 15th to 19th of July 2013 held at the Berlin State Library (in the following: Olszowy-Schlanger. "Name of the Lecture". ppt.).

ה)

–̣ used for <i>qamaš</i>	v. 7.3 העם for העם; v.7.3 וחרד for וחרד; (v. 7.9 יהוה ¹⁴⁰ for יהוה).
–̣ used for <i>šere</i>	v. 6.19 ויצא for ויצא; v. 7.3 וחרד for וחרד.
–̣ used for <i>ḥolam</i>	v. 6.19 ויצא for ויצא.

ו)

– ¹⁴¹ used for <i>'atnaḥ</i>	v. 7.2 בידם for בידם.
– ¹⁴² used for <i>ṭifḥa</i>	v. 7.2 מדין for מדין.
–̣ used for <i>zaqef qatan</i>	v. 7.2+3 לאמר for לאמר; (v. 7.9 יהוה ¹⁴³ for יהוה).
–̣ used for <i>tḥbir</i>	v. 6.19 אלי for אלי.
–̣ used for <i>rḥia'</i>	v. 7.3 ועתה ¹⁴⁴ (ועתה) for ועתה.
–̣ used for <i>pašṭa</i>	v. 7.2 ישראל for ישראל; v. 7.3 העם for העם; v. 7.9 ויהי for ויהי.

ז)

–̣ used for <i>'azla</i> ¹⁴⁵	v. 7.3 קרא for קרא.
–̣ used for <i>mahpak</i>	v. 7.2 עלי for עלי.
– ¹⁴⁶ general conjunctive	v. 7.2 מתתי את-מדין for מתתי את-מדין (<i>merṣka</i> –̣, and <i>maqṣef</i> –̣); v. 7.2 יתפאר עלי for יתפאר עלי (<i>'azla</i>).

140. Maybe to be read as *zaqef qatan*, since the same graphical sign used for both. However, I think it is more likely here that it represents *qamaš* due to the usual postpositiv position of *zaqef qatan* in this manuscript (see below).

141. Dot actually on baseline inbetween the letters. For the palaeographical terminology see Olszowy-Schlanger. Methodology Hebrew Palaeography. ppt).

142. Dot actually on baseline inbetween the lettres.

143. Maybe to be read as *qamaš*, since same graphical sign used.

144. According to Revell (1977, p. 37) it is a horizontal stroke.

145. The following world should have a gereš, but it is not legible.

146. Dot actually on headline between two words in the same place as *maqṣef* is usually marked.

ה) No other diacritical Palestinian signs used.

ו) The usage of Palestinian vowels cannot be classified according to Revell¹⁴⁷ due to the lack of evidence. Hypothetically, the available evidence points towards the first three or the seventh class.¹⁴⁸

ז) The single occurrence of a Tiberian grapheme shows Standard Tiberian pronunciation tradition, although this is not at all representative.

ח"ו) Revell's classifies this manuscript (P205) as belonging to group one.¹⁴⁹ He sees the vertical stroke of *zaqef* and horizontal stroke of *reḥia* ' as "distinguishing characteristics of this group".¹⁵⁰ Furthermore, he states that the accent signs tend to be marked on the last syllable in this group.¹⁵¹ This applies also for this manuscript. However, I do not agree that the "horizontal shaped" signs tend to be marked postpositive.¹⁵² In this manuscript only two (*reḥia* ' and *zaqef qatan*) out of five accents are marked in a postpositive positions. Moreover, only *reḥia* ' is in a "horizontal shape". Unlike in the Tiberian system, where these two accents are marked on the first letter of the stressed syllable, they are marked postpositive in this Palestinian tradition. Therefore, it is likely that in v. 7.9 the vertical stroke is used to designate a *qamaṣ* and not *zaqef qatan*. *Pašta* on the other hand is marked on the stressed syllable in this Palestinian system, but has a postpositive position in the Standard Tiberian tradition. There seems to be no connection between the positioning of the stress on the baseline and below or the headline and above, and the concept of מלעיל (*mil'el* "from above"(stress on the penultima)) and מלרע (*milra* ' "from below" (stress on the ultima)), since they all mark the stress on the last syllable (מלרע).¹⁵³

ט"ב) The word יצא (v. 6.19) is written defective, whereas the BHS has it *plene*. In v. 7.2 the

147. Revell. 1970. 111 (Table 2).

148. Revell. 1970. 111.

149. Revell. 1977. 18 and 37.

150. Revell. 1977. 37.

151. Revell. 1977. 37.

152. Revell. 1977. 37.

153. Joüon, P. et Muraoka, T. 2009. (2nd ed.). *A Grammar of Biblical Hebrew*. Gregorian and Biblical Press: Rom. 57.

word **יִהְיֶה**, according to the BHS, is written without **ה** as **יִפֹּאֵר**. There is no other diacritical sign marked, that would indicate an assimilation or a different stem. However, this lack of a diacritical sign is not unusual. In the *masorah parva* of the Aleppo Codex it has been noted, that **יִהְיֶה** occurs only once (ל).¹⁵⁴ Though, this refers only to the vocalisation, since **יִהְיֶה** can be found a second time at the end of a vers in pausal form (Is 44.23 **יִהְיֶה**). This may indicate that the scribe of the here treated manuscript did not come from the same masoretic tradition.

י"ג) The Palestinian vocalisation in this small manuscript differentiates between three vowel qualities in the same way, in which the Standard Tiberian tradition also does. However, this must not lead to the conclusion that the scribe was influenced by the Standard Tiberian pronunciation tradition, since the evidence is only scarce. Furthermore, the use of two conjunctive accent signs with the same function and shape as their Tiberian counterparts does also not point towards Tiberian influence, since it is not clear whether they were first employed in the Palestinian or the Tiberian system. Revell's classification system is applicable only in a limited way due to two reasons 1.) the classification of a manuscript according to its vocalisation requires the full set of vowel graphemes, 2.) Revell's classes of vocalisation¹⁵⁵ and accentuation¹⁵⁶ do not necessarily correspond. Thus, for this manuscript the first three classes could be appropriate. However, the use of the vertical stroke for *pašṭa* is only found in the classes three and seven.¹⁵⁷ This leads to the conclusion that the manuscript has to be placed in class three, since this is the only coinciding class for vocalisation and accentuation. Though, this is highly hypothetical due to the lack of other examples of vocalisation in this manuscript, which could prove the thesis.

4.4. T-S A 19.11

א) Daniel 6.27-10.17.

ב) Hebrew and Aramaic. Palestinian accentuation and vocalisation. Tiberian accentuation

154. Cf. Aleppo Codex (online via <http://www.aleppocodex.org/> [accessed at the 29.07.13]).

155. Revell. 1970. 111.

156. Revell. 1970. 97-98.

157. Revell. 1970. 97.

and vocalisation. Consonantal text, verse division and Palestinian vocalisation, accentuation in brown ink. Tiberian graphemes and maybe some different Palestinian graphemes in black ink (cf. below).

ג) Parchment; 16,8 x 25,9 cm and 17,3 x 27,1 cm;¹⁵⁸ 4 leaves (two bifolia); 1 column; 22-25 lines; mutilated with a few pieces of text missing on the corners. Oriental square script of the South-Western type (cf. ג of AS 63.6).

ד) No words with double vocalisation but Palestinian and Tiberian vowels and accents on different words.

ה)

◌ֿ used for <i>ševa mobile</i>	v. 8.12 בַּפֶּשַׁע for בִּפְשַׁע.
◌ֿ used for <i>hireq</i>	v. 8.7 רֵאִיתִיו for רֵאִיתִיו.
◌ֿ used for <i>šuruq</i>	v. 9.13 לָשׁוּב for לָשׁוּב.

ו)

◌ֿ used for <i>zaqef qatan</i>	v. 8.8 הַגְדֹּלָה for הַגְדֹּלָה.
◌ֿ ¹⁵⁹ used for <i>tiḥfa</i>	v. 8.20 רֵאִיתִי for רֵאִיתִי. Alternative: v. 8.1 לְמַלְכוּת for לְמַלְכוּת.
◌ֿ used for <i>pašta</i>	v. 8.21 הַגְדֹּלָה for הַגְדֹּלָה; v. 8.8 אֶרֶץ for אֶרֶץ; (e.g. addenda).

ז) No conjunctive accent signs clearly identifiable.

ח) No other diacritical Palestinian signs used.

ט) This manuscript does theoretically not fit into Revell's classification scheme, since there is no class that uses the Palestinian *šere* sign for Tiberian *hireq*.¹⁶⁰ Yet, Revell gives

158. Davis. 1978. 46 (vol. 1); Davis/Outwaite. 2003. 492 (vol. 4). Reference to Revell (1977) see below. This manuscript is also mentioned several times in Chiesa, B. 1978. *L'Antico Testamento Ebraico secondo la tradizione palestinese*. Bottega d'Erasmus: Torino to which I did not have access.

159. Dot actually on headline after the word.

160. Revell. 1970. 111 (Table 2).

multiple examples for this phenomenon in his explanation to the classification.¹⁶¹ However, he does not regard this as a distinctive feature for his classification scheme.¹⁶²

י) In this manuscript the *rafe* sign is used on some words and always corresponds to Standard Tiberian tradition e.g. v. 6.29 בְּמִלְכֻתָּהּ. Tiberian graphemes are employed only seldom and their usage varies. Thus, one finds traces of the Predominately and the Non-Standard Tiberian pronunciation traditions e.g. in v. 6.28 where *pataḥ* is changed to *qamaṣ* i.e. מֶן יָד instead of מִן יָד.¹⁶³ Since this verse of Daniel is in Biblical Aramaic, the interchange with *qamaṣ* seems to show Hebrew influence. In Biblical Aramaic both states of יָד i.e. construct state and absolute state are written with a *pataḥ* in the first syllable, if they do not take a suffix.¹⁶⁴ Though, if this form was in ordinary Biblical Hebrew, i.e. the Tiberian Standard tradition, then it would still be written with a *pataḥ* in the first syllable, since it is the *nomen regens* of a construct state.¹⁶⁵ The vocalisation of the word following מֶן יָד is also noticeable. In the Leningrad Codex it is vocalised as אֲרִי־יָהּ¹⁶⁶, whereas it is here vocalised as אֲרִי־יָהּ. The

161. E.g. Class 1: T-S NS 249.5: (י) לִי in Ezek 35.10 (Revell. 1970. 75); class 12: T-S 12.197 סִיר in Jer 1.13 (Revell. 1970. 95).

162. "However, if only the broad characteristics of each ms are taken into account [...] it will be seen that the differences in the mss are well defined. These differences, which are most clearly shown in the use of the 'a' and 'e' signs, can serve as a basis for a classification into which the pointing of any known ms will fit." (Revell. 1970. 32).

163. The masoretic note to this combination states: יוּ בִּיחַד מֶן, which can be translated as "16 times united with מֶן. At first sight, this remark seems misplaced, since in this case it is not connected. The *sefer 'okhlah we'okhlah* clarifies this by elaborating that this is the only occurrence of this combination where it is not connected (Frensdorff, S. 1864. *Das Buch Ochlah W'ochlah (Massorah)*. Hahn'sche Hofbuchhandlung: Hannover. 122 (Nr. 196)).

164. Bauer, H. et Leander, P. 1969. *Grammatik des Biblisch-Aramäischen*. Olms: Hildesheim/ New York. 214.

165. Joüon/Muraoka. 2009. 253 (§92). E.g. Lev 25.47 :[...] וְיָדָהּ יָדָהּ.

166. The etymological origins of this word are not yet unearthed. However, the following thesis regarding the phonological development has been published by Bauer/Leander (1969, 193 (§51m")). They suggest this is the plural form of אֲרִיָּהּ and the original form might have been *'aryay. Evidence that אֲרִי־יָהּ is not an exceptional vocalisation pattern of Biblical Aramaic can also be found in other texts. For example, in a Talmudic manuscript with Babylonian vocalisation the form אֲרִי־יָהּ (Sokoloff, M. 2002. *A Dictionary of Jewish Babylonian Aramaic*. Bar Ilan UP/The Johns Hopkins UP: Ramat-Gan/ Baltimore and London. 166) is delivered. Furthermore, in Mandaic אֲרִי־יָהּ is handed down, in which א has been inserted for an [a] vowel (Sokoloff. 2002. 266; Drower, E. S. et Macuch, R. 1963. *A Mandaic Dictionary*. Clarendon Press: Oxford. 37. This is also attested in the Karaite Hebrew Bible manuscripts transcribed into Arabic (cf. Khan, G. 1993. "The Orthography of Karaite Hebrew Bible Manuscripts in Arabic Transcription". *Journal of Semitic Studies* 38. 59-70). Since Mandaic and Jewish Babylonian Aramaic share some similarities this could explain the similar reading tradition (cf. Burtea, B. 2011. "Mandaic". S. Weninger (ed.), G. Khan et al. *The Semitic Languages. An International Handbook*. Berlin and Boston: De Gruyter Mouton. 671). Furthermore, in Targumim with Babylonian vocalisation forms

ending -ות cannot be explained phonologically, i.e. with the strong tendency of monophthongization in Biblical Aramaic.¹⁶⁷ Therefore, it has to be explained morphologically. I think the *naqdan* only knew the Hebrew form and therefore vocalised it accordingly. Besides the final א this form has the same consonantal pattern as its Hebrew cognate i.e. אריות¹⁶⁸. In v. 7.16 the vocalisation of קאמנא is changed to קאמנא. The change from *ḥaṭef pataḥ* to *pataḥ* could indicate that their quantitative differences were not perceived as such or that, like in the Palestinian system, there was no need to indicate these differences. The change from *pataḥ* to *qamaṣ* points towards the Sephardic pronunciation tradition.¹⁶⁹ In this pronunciation tradition no distinction is usually¹⁷⁰ made between the spelling of long *qamaṣ* (*gadol*) and *pataḥ*.¹⁷¹ Concluding one can say, that in this case it seems as if *pataḥ* was used to represent short vowels and *qamaṣ* to represent long vowels. According to the Standard Tiberian rules both of them could be long or short depending on the syllable

like אריות and אריות (Sokoloff. 2002. 166) are attested. I think that in Hebrew and Aramaic the forms underwent different developments. Thus, in Aramaic the origins are only preserved in the plural, whereas in Hebrew and the Aramaic singular the "new" developed forms אריות and אריות were used to which only the plural ending has been added (in Hebrew mostly feminine but once masculine (1Kings 10.20 אריות)). However, the pattern of אריות, which seems to be the same pattern as that of מלכותא (Gzella, H. 2011. "Imperial Aramaic". S. Weninger (ed.), G. Khan et al. *The Semitic Languages. An International Handbook*. Berlin and Boston: De Gruyter Mouton. 578.), raises questions. For example, why has *šva mobile* changed to long *qamaṣ* (Bauer/Leander. 1969. 234 (§62i); Gzella mentions only that words of this pattern tend towards triphthong (/awātā/) endings with a short vowel, but he gives no explanation for this change (2011, 578)). Whether this could be explained by the insertion of an epenthetic vowel has to be evaluated more carefully on similar examples from the Babylonian pronunciation tradition and Aramaic (cf. Khan, G. 2013. "Epenthesis: Biblical Hebrew". G. Khan (ed.), S. Bolozky et al. *Encyclopedia of Hebrew Language and Linguistics*. Brill: Leiden. (forth.)).

167. Bauer/Leander. 1969. 38 (§9p): "aṯ wurde im Bibl.-aram. immer zu ō assimiliert [...]".

168. E.g. Judg 14.5; 2Sam 1.23; 1Kings 7.29.

169. Cf. e.g. Blau. 2010. 108 (§3.5.3.1) "In the accepted Sephardic pronunciation (originally) short and long *qamaṣ* are pronounced differently. The short vowel, originating in *u*, as *o*, and the long vowel, deriving from (original or secondary) *ā* as *ā*." Blau. 2010. 112 (3.5.6.1.1n) "As noted above, *a* reflects the Sephardic pronunciation of mobile *šwa*. According to Tiberian tradition it is, in the main, pronounced as *ā*, thus identical to *ḥaṭef pataḥ*." cf. e.g. also Khan. 2013. *Biblical Hebrew: Pronunciation Traditions*; and Khan, G. 2013. "Shewa: Pre-Modern". G. Khan (ed.), S. Bolozky et al. *Encyclopedia of Hebrew Language and Linguistics*. Brill: Leiden. (forth.).

170. Such a distinction can be found in the pronunciation tradition of Biblical Hebrew of Iranian languages speaking communities, although their pronunciation is classified as Sephardi. Iranian vernaculars distinguish between the quality of short and long [a] vowels. Thus, long *qamaṣ* is pronounced with a back quality [a:] or [ɔ:] and *pataḥ* with the front quality [a]. Furthermore, Khan believes that this pronunciation tradition influenced other Hebrew reading traditions (Khan. 2013. *Biblical Hebrew: Pronunciation Traditions*).

171. Blau. 2010. 108 (§3.5.3.1).

structure and the placement of stress.¹⁷² In v. 7.23 the word ¹⁷³וְתִקְנָה is noteworthy, since it is vocalised as וְתִקְנָה. *Seghol* for *pataḥ* is likely to reflect the Babylonian pronunciation tradition,¹⁷⁴ in which only one sign exists for both Tiberian vowels.¹⁷⁵ The change from *ševa mobile* to *pataḥ* could indicate a phonetic realisation like that of Standard Tiberian as short [a] sound.¹⁷⁶ In v. 7.25 יִמְלֵל becomes יִמְלֵל. This phenomenon can be explained with the historical development from short [i] to short [e] in Aramaic.¹⁷⁷ While still in progress in the time the Hebrew Bible was written, this development has been completed by the time this manuscript was copied.¹⁷⁸ Therefore, one can conclude that short [i] in this position reflects a historical pronunciation of Aramaic from the Second Temple period. In v. 7.28 רָעִיוֹנִי becomes רָעִיוֹנִי. This reflects the original Standard Tiberian pronunciation tradition that reads *ševa mobile* before *yod* as short *hireq* (*rā'iyōnāy*).¹⁷⁹ However, it does not reflect the vowel notation of Standard Tiberian.¹⁸⁰ To the word וְהַרְמִסָּם at the end of v. 8.10 a *yod* is added וְהַרְמִסָּם. This seems to reflect secondary lengthening effected by *minor ga'ya*.¹⁸¹ In v. 8.17 הֵבִין is written with the second radical *yod* הֵבִין.¹⁸² The general tendency of Hebrew to mark vowels

172. Khan, G. 2013. "Vowel Length (i) Biblical Hebrew". G. Khan (ed.), S. Bolozy et al. *Encyclopedia of Hebrew Language and Linguistics*. Brill: Leiden. (forth.).

173. The *masorah parva* to this word states that this is also a *hapax legomenon* (ל).

174. Khan. 2013. Biblical Hebrew: Pronunciation Traditions.

175. Dotan, A. 2007. "Masorah". M. Berenbaum et F. Skolnik (eds.). *Encyclopaedia Judaica, Vol. 13 (2nd ed.)*. Macmillan Reference: Detroit. 630 (§5.2.1.1.1). According to Yeivin, a sign, that looks like the Palestinian sign for *seghol* ְ, was sometimes used in the Babylonian pronunciation tradition to indicate short [a] (Yeivin, I. 1985. *The Hebrew Language Tradition as Reflected in the Babylonian Vocalization*. The Academy of the Hebrew Language: Jerusalem. 365-366. [Hebrew]).

176. Dotan. 2007. 631 (§5.2.1.1.1). However, Khan (2013, Shewa: Pre-Modern) states that the phonetic realisation of Babylonian *ševa mobile* is not clear. For the pronunciation of Tiberian *ševa mobile* cf. Khan. 2013. Shewa: Pre-Modern and Dotan. 2007. 633 (§5.3.1.2).

177. Bauer/Leander. 1969. 30-31 (§6b'-c'); 110 (§35b). Cf. for the description of the same development in Biblical Hebrew Blau. 2010. 133 (§3.5.8.2).

178. Bauer/Leander. 1969. 30 (§6c'). According to Bauer/Leander this is reflected in Jewish-Palestinian Aramaic (1969, 30 (§6c')).

179. Khan. 2013. Shewa: Pre-Modern.

180. Though, it reflects the vowel notation according to ben Naftali.

181. Khan. 2013. Vowel Length (i) Biblical Hebrew.

182. I think it is unlikely that the *munah* (conjunctive accent) effected a lengthening of the vowel, since only strong disjunctive accents or secondary stress can evoke such a lengthening (Blau. 2010. 154 (§3.5.13.1); Khan. 2013. Vowel Length (i) Biblical Hebrew).

with their corresponding semi-vowels could serve as an explanation. However, this form could also show that it was derived from the "regular" prefix tense (יָקִים) and not from the short prefix tense (יָקֵם).¹⁸³ According to Blau the imperative is usually represented by the pattern of הָקֵם.¹⁸⁴ In the margin of v. 9.13 the word אֲבוֹתֵינוּ is written in black ink. It was omitted by the scribe that wrote the consonantal text. In comparison to the BHS vocalisation of this word (אֲבוֹתֵינוּ) two features are noteworthy. The change of *ḥatef pataḥ* to *pataḥ* and the *plene* writing of *ḥolam*. The first phenomenon has already been described above and might reflect a non-awareness of ultrashort vowels. The second phenomenon might mark a tendency towards general plene spellings (see above הָבִין). In this manuscript not only Tiberian vowel signs are used but also Tiberian accent signs. Thus, for example, in v. 7.4 *munah* on קְדַמִּיתָא, in 7.5 *ḥir* on עֲלֵעִין and finally in v. 7.3 *zaqef* on רַבְרָבִין are used in the same way as in the BHS.

(י"א) According to Revell the pattern of the accent signs of this manuscript belongs to group I, C.¹⁸⁵ This is due to the graphic signs that represent *zaqef* and *reḥia*.¹⁸⁶ However, according to my findings in this manuscript does *zaqef* differ from his observation regarding the shape, which Revell describes as a vertical stroke. However, I found a inscribed postpositive dot. Furthermore, for *ṭifḥa* he gives examples of a subscribed dot, which seems to have ascended to the headline in this manuscript. He concludes that *pašṭa* is written "[...] as a guide to the accentuation of the *zaqef* clauses [...]."¹⁸⁷ To a certain extent, I can confirm this observation due to regular occurrences of *pašṭa*. However, it remains a theory, since in this manuscript they are not clearly identifiable due to the condition of the ink and the parchment.¹⁸⁸

(י"ב) Although the Tiberian *rafe'* sign used (e.g. v. 6.28 יָד), no *dageš* sign can be found.

183. Blau. 2010. 308. Blau thinks there is no evidence for this form, therefore, he classifies it as non-existent (2010, 308).

184. Blau. 2010. 308.

185. Revell. 1977. 29. For details of this class see Revell. 1977. 55-56.

186. Revell. 1977. 55.

187. Revell. 1977. 56.

188. Cf. my notes to this manuscript in the appendix.

added word P2 (B) in the margin with black ink. In v. 7.23 the word ¹⁸⁹וְהָאֵל is copied and vocalised as וְהָאֵל. The omittance of א points towards the loss of its consonantal value and the lack of knowledge about the orthography of this root and form.¹⁹⁰ The same phenomenon occurs in v. 7.15 where the final *aleph* of בָּנוּא is omitted and בָּנוּ is read.¹⁹¹ In the same verse it seems as if *seghol* looks like an actually non-existent *ḥatef ṣere* (נִרְנָה), sharing the right dot of *ṣere* and the upper dot of *ševa* in a right angle. In v. 8.1. *tifḥa* is marked on לַמַּלְכוּת with black ink against other examples of accentuation signs that are written in the ink of the consonantal text. The verse division in this manuscript is regularly marked according to the BHS. Furthermore, also paragraphs are indicated where BHS has פ (for פְּתוּחָה)¹⁹² (v. 7.1; 7.14; 8.1; 9.1 (see below); 10.1) but their layout does not follow the rules of Maimonides, as, for example, the Leningrad Codex does.¹⁹³ Instead of switching to a new line or leaving one blank the scribe has just left space of about eight letters between the ending and starting of two paragraphs. Moreover, two decorated ס can be found in the margins, that mark the division into *sedarim* of the Palestinian reading cycle.¹⁹⁴ Thus, at the beginning of the line where v. 6.29 ends and verse 7 starts and also where v. 8.27 ends and verse 9 begins. However, the second occurrence does not correspond to the BHS,¹⁹⁵ since the *seder* starts here at v. 9.4 and not v. 9.1. In v. 8.13 קֹרֵשׁ has two vertical dots that most likely represent the Babylonian /o/ sound.¹⁹⁶ However, this is just a vague suggestion, since the upper dot is in black ink and the lower one in brown ink. Furthermore, the conjunction ו is here omitted. This might be a mistake due to the two preceding קֹרֵשׁ in this verse without initial ו or it is just a variant.¹⁹⁷

189. The *masorah parva* to this word states that this is a *hapax legomenon* (ל).

190. Blau. 2010. 86 (§3.3.4.1). Here especially Blau. 2010. 88-89 (§3.3.4.3.1-2).

191. Blau. 2010. 86 (§3.3.4.1).

192. Yeivin. 1980. 40-41 (§74).

193. Yeivin. 1980. 40-41 (§74). Cf. also Khan. 2012. 35-36.

194. Yeivin. 1980. 39-40 (§72).

195. Here the BHS is correctly displaying B19a.

196. Dotan. 2007. 631 (§5.2.1.1.1).

197. Cf. The variants between ben Asher and ben Naftali (Yeivin. 1980. 142 (§155-156)) elaborated in the *sefer ha-ḥillufim* by Misha'el ben 'Uzziel (10th century CE). One of the examples given by Yeivin deals also

י"ג) *ševa mobile* is here marked with the grapheme that is supposed to represent *seghol*.¹⁹⁸ Unlike the Tiberian system the Palestinian system has no particular grapheme to mark *ševa* or other *hatefim* vowels. Therefore, vowel graphemes are borrowed, which represent in other instances full vowels.¹⁹⁹ However, most commonly used are signs for the [a] (i.e. *qamaš* and *pataḥ*), [e] (i.e. *šere* and *segol*) and [i] (i.e. *hireq*)²⁰⁰ sounds.²⁰¹ Moreover, out of these *šere* and *segol* are employed most frequently.²⁰² This tendency points towards the Sephardic pronunciation tradition, in which *ševa mobile* is pronounced as [e] sound.²⁰³ According to Revell [a] sounds are only used under specific circumstances.²⁰⁴ The two horizontal dots on the ש of לְשׁוֹב do not necessarily mark an [u] sound, since the same sign is also employed to represent the Palestinian *pašta*. Unlike in the Tiberian system, this accent is not necessarily placed in postpositive position (cf. T-S AS 63.6). The [e] grapheme on רִאִיתִי seems to be added in a different ink (black), which leads to the conclusion that at least two *naqdanim* have been working on this manuscript and both of them knew the Palestinian system.²⁰⁵ Summarizing the vocalisation findings of this manuscript one can conclude that the Tiberian vocalisation shows some Palestinian (i.e. Sephardic) influence. However, it does still show characteristics of Standard Tiberian, which do not reflect its proper orthography but they reflect its correct spelling.

with a similar problem: Jer 7.25 עָד (ben Asher) or עָדָה (ben Naftali) (Yeivin. 1980. 142 (§156)).

198. Revell. 1970. 10; Dietrich. 1968. 87*-88*; Kahle. 1901. 300.

199. Revell. 1970. 86-87. Dietrich (1958, 117-121; especially 117, note 2).

200. This occurs only where *ševa* is followed by yod (e.g. Revell. 1970. 81). This feature is also known from the Aleppo Codex and especially the so-called ben Naftali masoretic school (Yeivin. 1980. 142 (§156)). Furthermore, it is also a topic of discussion in the *sefer ha-ḥillufim* which treats the differences between Ben-Asher and Ben-Naftali (Yeivin. 1980. 141-143 (§155-156)). Thus, it represents not a particular Palestinian tradition, since it is also found in the Standard Tiberian tradition (Khan. 2013. Shewa: Pre-Modern).

201. Revell finds all vowel signs used for *ševa mobile* in Biblical texts (cf. e.g. 1970, 86-87), whereas Dietrich only notes the usage of [a] and [e] graphemes (1958, 119).

202. Dietrich. 1968. 87*-88*.

203. Khan. 2013. Shewa: Pre-Modern Hebrew; Blau. 2010. 119 (§3.5.6.7.2).

204. Revell. 1970. 86-87.

205. Cf. my notes in the appendix and Revell. 1977. 29.

4.5. T-S AS 63.59

א) Leviticus 19.15-20.3.

ב) Hebrew. Tiberian vocalisation with the Palestinian sign for [o] $\dot{\text{◌}}$ besides Tiberian *ḥolam* grapheme.²⁰⁶ Only Tiberian accentuation signs used. One type of ink used (brown).

ג) Parchment (yellow colour could point towards goat skin)²⁰⁷; 13.2 x 13.2 cm; 1 leaf; 1 column; 13 lines on *recto* and 14 lines on *verso*; mutilated and rubbed (especially *recto*) with pieces of text missing or illegible.²⁰⁸ Oriental square script of the South-Western type.

ד) $\dot{\text{◌}}$ and $\text{◌}^\text{◌}$ from אהו.

ה)

$\dot{\text{◌}}$ used for <i>ḥolam</i>	e.g. v. 19.15 ול $\dot{\text{◌}}$ (ולא) for ול ◌ ; v. 19.15 ו $\dot{\text{◌}}$ (נדול) for נדול;
--	--

ו) No disjunctive Palestinian accent signs used.

ז) No conjunctive Palestinian accent signs used.

ח) No diacritical Palestinian signs used.

ט) All the occurrences of the Palestinian [o] signs correspond only with Standard Tiberian *ḥolam* either *defective* or *plene*. I think, this manuscript does not reflect Palestinian pronunciation but rather a Tiberian.²⁰⁹ The Palestinian sign is in my opinion only used, because it is much clearer and easier to read than just the single dot of the Tiberian [o], especially, when no other letter is following this sign e.g. v. 19.26 ל $\dot{\text{◌}}$ for לא this can be proven by the fully written word that only use the Tiberian sign e.g. v. 19.18 תִּקְרָם and v. 19.26 תִּעֲנֶנִי.²¹⁰

206. Friedländer, M. 1896. "Some Fragments of the Hebrew Bible with Peculiar Abbreviations and Peculiar Signs for Vowels and Accents". *Proceedings of the Society of Biblical Archaeology* 26/3: 89.

207. Olszowy Schlanger. Italian Script.

208. Davis/Outhwaite. 2003. 414 (vol. 4).

209. Revell does not consider this manuscript as being of any interest, since in his opinion "the pointing of the MS is considerably closer to the Standard Tiberian than is that of many MSS which use only Tiberian signs." (1977, 3).

210. Friedländer suggests the Palestinian sign is used, so that the Tiberian *ḥolam* cannot be mistakingly taken for *reḥia* (Friedländer. 1896. 93).

However, the following examples are the only two exceptions for these two examples: v. 19.19 בֹּרֶךְ for בְּרָכָה of and 19.27 רָאִשְׁכֶּם.²¹¹

י) In this manuscript usually only the stressed syllables are written (see also ב"ב). Thus, not only words with disjunctive accents are written but also those with conjunctive accents and secondary stress. In *maqgef* constructions the word preceding and the word following the *maqgef* are indicated by at least writing one letter. This is usually the first letter of the last syllable of the preceding word and at least the stressed syllable of the following word e.g. v. 19.20 אֶשֶׁ for אֶת־אֶשֶׁ. Although there are no changes regarding the signs used for neither the vocalisation nor the accentuation in comparison to B19a, I would like to briefly write about an observation regarding the placement of stress. When an abbreviated word is accentuated with an usually postpositive accent (in this case *segholta*, *zaqqef* and *telisha parvum*) it is usually placed on the initial letter of the syllable and therefore is no longer postpositive e.g. v. 19.16 רָכַ for רָכִיל; v. 19.19 בֹּרֶךְ for בְּרָכָה; v. 20.2 אֶ for אִישׁ. However, if the whole word is written, it retains in its natural environment the postpositive position e.g. v. 19.32 אֶל־הָאֵבֶה; v. 19.32 שִׁבְּהָ; This leads to the conclusion that they only change position for the convenience of the abbreviated manuscript, since space can be saved by only writing the initial letter of a syllable. As I already have shown in the preceding manuscripts, the Palestinian accentuation tends to mark its *pašta* equivalent also not in postpositive position. Whether these two observations are connected, has to be proven with more detailed research on abbreviated manuscripts. Friedländer thinks "that in the mind of the copyist the third system was somehow connected with the abbreviations."²¹² Also three non-standard features of Tiberian vocalisation tradition occur in this manuscript. 1.) *Rafe* is also marked on other letters than the *BGDKFT* e.g. v. 19.30 וּמִקְרָשׁ for וּמִקְרָשִׁי and v. 19.28 הִתְהַנֵּנוּ²¹³ for הִתְהַנְנוּ; v. 19.20 שִׁבְּבָה²¹⁴. Although they are not used according to the rules of marking *rafe* in the Standard Tiberian²¹⁵

211. For more examples see the list with all occurrences in the appendix.

212. Friedländer. 1896. 94.

213. In my opinion, *rafe* is marked here to show that it is not a nun-energicum (Joüon/Muraoka. 2009. 160-161 (§61f)).

214. ב receives *rafe* to show that it is pronounced soft as labio-dental fricative [v] (Khan. 2012. 73-74).

215. It is, for example, not coherently used in B19a e.g. Lev 19.37 הִקְטִי (cf. on *rafe* Blau. 2010. 78-79 (§3.3.2.1)).

tradition, they still reflect this pronunciation tradition, since firstly, the vocalisation has not been changed on any instance and secondly, I think they are not only employed to give details about the pronunciation of the letters, which is the case on *BGDKFT* letters, but also to indicate that neither *dageš* nor *mappiq* is used, which is the case with other than the *BGDKFT* letters.²¹⁶ 2.) *Dageš* (most likely as *lene* rather than *forte*)²¹⁷ is not only used on the *BGDKFT* letters but also on ז as in e.g. v. 19.33 בְּאַרְצָם²¹⁸ and several times in ל of לֹא as in e.g. v. 19.15 לֹא־.²¹⁹ For the latter usage of *dageš* no rules can be established, since I cannot find any pattern in the syllable structure of the preceding syllable. Thus, I think, like Morag already pointed out, that *dageš* is used as a syllable-boundary marker according to the Standard Tiberian rules that are applied for the *BGDKFT* letters.²²⁰ However, this is not a feature of Standard Tiberian, but it already occurs in B19a e.g. Lev 20.1 לֹא־נָרָה.²²¹ Finally the 3.) feature is the usage of the sign for *ševa* to indicate that the final consonant of a word is vowelless i.e. *ševa quiescens* as in v. 19.27 בְּאַרְצָם and 19.30 בְּאַרְצָם.²²² However, this already occurs in Standard Tiberian e.g. 2Sam 6.22 אֲבִירָה (i.e. the suffix-tense 2f. sg.).²²³ Although, these graphic features

216. It has to be mentioned that already in B19a *rafe* is used to show that א is silent e.g. Lev 19.27 רָאִשָׁם. This is not indicated in the BHS nor in the new BHQ.

217. Cf. Morag, S. "The Vocalization of Codex Reuchlinianus: Is the "Pre-Masoretic" Bible Pre-Masoretic?". *Journal of Semitic Studies* 4/3:226 (§2.36).

218. Here *dageš* shows that the preceding syllable is closed.

219. Only ל of לֹא are affected. However, not all of them are marked (Friedländer thinks that *dageš* is used in order to emphasise the letter (1896, 89-90) with which I agree). This non-standard *dageš* comes with another remarkable feature. It seems as if some of these לֹא, but not לֹא, that precede a *maqṣef* construct receive secondary stress i.e. *ga'ya* (Dotan. 2007. 641 (§5.3.3.3.1.2 (b))) because it has transferred its main stress to the following word due to *maqṣef*, which can be seen as a feature of Standard Tiberian. However, it appears not in all instances in codices like B19a, since it is not marked consistently there.

220. Morag. 1959. 226 (§2.36).

221. Cf. Golinets, V. 2007. [Review]. "Biblia Hebraica quinta editione cum apparatu critico novis curis elaborato. 18. General Introduction and Megillot". *Orientalistische Literaturzeitung* 102/4-5:499, who found out, that some *dageš* in the BHS are actually flecks in the original manuscript of B19a. This interchange happened, since the BHS resp. BHK were produced based on black and white images of B19a, on which it is not possible to distinguish between the dots made by ink and the natural ones. However, it is still hard to distinguish them in coloured and zoomable images today.

222. These are the only two occurrences.

223. Khan. 2013. Epenthesis: Biblical Hebrew. Usually, the sources (Ibn Janah) state that the second *ševa* is silent. However, there are also sources (David Qimhi) that find this *ševa* being vocalic (Khan, G. 1996. "The Tiberian Pronunciation Tradition of Biblical Hebrew". *Zeitschrift für Althebraistik* 9:16 (n. 91); Khan. 2013. Syllable Structure: Biblical Hebrew).

do not reflect Standard Tiberian vocalisation tradition, they represent the Standard Tiberian pronunciation tradition. Thus, they are just used in an extended way in order to mark more clearly Standard Tiberian vocalisation tradition. Unlike Codex Reuchlinianus²²⁴ there are no changes in vocalisation, but the manuscript shows a similar tendency of marking *rafe*, *dageš* and *silent ševa* at the end of a word.

א"י) No Palestinian accentuation signs used (cf. י and י).

ב"י) In this manuscript every word is indicated usually by one letter or sometimes by two or three letters.²²⁵ However, all of the them bear the stress of the words they are representing. This is in all cases corresponding with B19a (For the position of the accentuation sign see above). Though, I cannot find a distinctive feature regarding the words abbreviated with one letter and those with two ore more letters. Thus, if a particular punctuation feature is especially interesting the scribe includes this by writing more than one letter. Another noteworthy feature is the abbreviation of the tetragrammaton. In this manuscript it is abbreviated as י, which, in my opinion, points towards the reading as ידני.²²⁶ A further feature of this manuscript is, that it has Masoretic notes in the margins, which are no longer fully legible. They are also indicated by the circle as in B19a.

ג"י) This manuscript shows some remarkable features regarding the connection between the Palestinian and Tiberian system. Moreover, I think I have demonstrated that it is important to study also such manuscripts that at first sight are not very promising. For example, it seems now evident, that abbreviated manuscripts with different vocalisation systems share some features, like the non-postpositive placement of *pašṭa*. The small differences, in its literally meaning, like *dageš* exemplify that this manuscript shows a development regarding the tradition of vowel notation in comparison to B19a, though its common roots are still visible.

224. Morag. 1959. 230 (§3.3).

225. At the beginning of a new verse the first two words are usually written fully. For shorthand-writing of manuscripts with Babylonian vocalisation see Yeivin, I. 1962. "A Babylonian Fragment of the Bible in the Abbreviated System". *Textus* 2:120-139. For more bibliographical references see the excursus on shorthand-writing.

226. Yeivin (1962, 122) shows that in abbreviated manuscripts with Babylonian vocalisation the tetragrammaton is represented by ידני.

5. Excursus Shorthand

In the following I will be giving a brief introduction to the research history of shorthand writing.²²⁷ According to Neubauer it is already mentioned in the Talmud.²²⁸ The word נוטריון is employed there to refer to this particular style of writing.²²⁹ It has been suggested being a Latin or Greek word despite the fact that it does not occur in an dictionary.²³⁰ However, I think it could be derived from the Latin *notarius*, who was skilled in writing in shorthand.²³¹ Furthermore, Neubauer in accordance with Krauss states that the Rabbis accepted the Roman way shorthand writing.²³² Primary resources are the Babylonian (Giṭṭin 60b) and Palestinian Talmud (Megillah 73a).²³³ The latter is referring to it regarding to the scroll of Esther, that can be used in shorthand writing for synagogal reading.²³⁴ In the Babylonian Talmud one passage deals explicitly with this issue. The question is there whether it is allowed to write only portions of the Torah. This is supported by the fact that Queen Helene donated a golden tablet to the Temple, on which the *soṭah*²³⁵ passage was inscribed and from which the Kohen copies. In the following discussion the question arises how the text is written on this table.

bGiṭṭin 60a

[...] אמר רבי שמעון בן לקיש משום ר' ינאי באל"ף ב"ת [...] הבא במאי עסקינן בסירונין²³⁷ [...]]

"Rabbi Shimon ben Laqish said in the name of Rabbi Jannai [the *soṭah* passage is written in] Aleph Bet [i.e. is only the first letter of each word] [...] What are we dealing with here ? With

227. For practical evidence see T-S AS 63.59 and my description above and my notes in the appendix.

228. Neubauer. 1895. 62.

229. Friedländer (1896, 86) notes one occurrence for this term: Mishnah Shabbath, 12,5.

230. Neubauer. 1895. *ibid*.

231. Meyer. N. N. "Notary (CT)." *Brill's New Pauly*. H. Cancik et H. Schneider (eds.). Brill Online. 2013. [accessed 21.08.13].

232. Neubauer. 1895. 362.

233. Neubauer. *ibid*. For more information see "Abbreviations." *Brill's New Pauly*. H. Cancik et H. Schneider (eds.). Brill Online. 2013. [accessed 21.08.13].

234. Neubauer. 1895. 362-363. The "Hebrew" word used in this passage is גינוסיון, which is identified as either Latin *cognitum* or Greek γινωσκον, that could be derived from γινωσκειν (Neubauer. 1895. 363).

235. Num 5.11-31.

236. Stevenson, W. B. 1924. *A Grammar of Palestinian Jewish Aramaic*. Clarendon Press:Oxford. 55.

237. Jastrow. 2004. 987.

interruptions [i.e. words that are abbreviated]."²³⁸

The manuscripts, which Neubauer is describing in his article, are written with Palestinian vocalisation, but it is not his aim to talk about the vocalisation.²³⁹ I think by then, Neubauer thought that these unknown dots belonged to the abbreviation system. However, this was not yet known at this point, since only the Babylonian and Tiberian systems were known. Friedländer was the first to pay attention to the vocalisation of these manuscripts.²⁴⁰ He examines the vocalisation signs in detail and gives description of other features found in abbreviated manuscripts. However, his article from 1896 remains until today, as far as I know, the only attempt to describe not only the vocalisation but also the abbreviations. He notes three classes of abbreviated manuscripts: 1.) Fragments with Palestinian vocalisation and accentuation, 2.) Fragments with Tiberian vocalisation and accentuation, which in certain cases use the Palestinian *ḥolam* (as shown above in T-S AS 63.59), 3.) Fragments with Tiberian vocalisation and accentuation.²⁴¹ Paul Kahle continues with publishing studies on shorthand manuscripts.²⁴² He introduces the term סירויין and uses it regularly afterwards.²⁴³ However, he is focussing on the vocalisation and accentuation. To my knowledge, Yeivin was the last scholar to publish an article on shorthand. In his article he deals with a manuscript with Babylonian vocalisation.²⁴⁴ I think that a continuation and careful examination of abbreviated manuscripts with different vocalisations could shed light on differences and similarities not only regarding the vocalisation traditions but also regarding the abbreviation systems.

238. Isbee, Y. et Kuber, M. (eds.). 1993. "מסכת גיטין" (Tractate Gittin). Volume 2". H. Goldwurm et al. (eds.), תלמוד בבלי (Talmud Bavli. The Schottenstein Edition). Mesorah Publications: Brooklyn. 60a³. See also bYoma 37a, which deals with this particular tablet.

239. Neubauer. 1895. 361-364.

240. Friedländer. 1895. 86-98.

241. Yeivin. 1962. 120; Friedländer. 1896. 86-98.

242. Kahle. 1901. 273-317; Kahle, P. 1967 (Repr. 1927). *Masoreten des Westens* (vol. 1). Olms: Hildesheim. 25; Kahle, P. 1967 (Repr. 1927). *Masoreten des Westens* (vol. 2). Olms: Hildesheim. 31*-35*.

243. Kahle, 1901. 275. However, already Friedländer (1896, 87) mentions this word, but he does not use it as a term.

244. Yeivin. 1962. 120-139.

6. Conclusion

The decision to include a part for the classification according to Revell's classes in the examination scheme was not really satisfying, although it served as a good background to explain certain features. This was mainly due to the fact that none of the manuscripts which I examined were using a full set of vowels. Although, I find Revell's approach appropriate and sophisticated, I found it very difficult to follow his classes. In order to elaborate a classification scheme he had to neglect a lot of other evidence from the respective manuscripts. Although, for example, in group one of his classification no interchange between *seghol* and *šere* occurs,²⁴⁵ both signs are used interchangeably. Moreover, also [a] and [e] signs are used interchangeably.²⁴⁶ Although, these are just minor details, they are important when it comes to classification. Revell values in this manuscript only that four vowel graphemes for [e] and [a] sounds are used mostly in accordance with Standard Tiberian, even though he notes the other characteristics and states this distinction.²⁴⁷ In my opinion, the distinctive features that I mentioned above should be valued more, since they can point towards other influence too i.e in this case the Babylonian vocalisation.²⁴⁸

As I have shown, evidence of Palestinian vocalisation is very scarce not only in my manuscripts but also in those that have been examined by Revell and others. Sometimes only one syllable in a word is marked. Thus, the scholar relies on the vocalisation signs marked by the scribe, which he chooses to note. Therefore, it is possible that this represents only certain cases and does not give an accurate picture.

This scarce notation of vowels could lead to the suggestion that shorthand-writing is connected with the Palestinian vocalisation. Although, this cannot be undoubtedly proven, there seems to be a connection. It appears to me that both come from the same idea of serving as reading aids. However, all inventions of vocalisation systems go back to this idea, but unlike manuscripts with Babylonian and Tiberian systems, which can be found fully vocalised, I have not read about finds of fully vocalised manuscripts with the Palestinian

245. Revell. 1970. 111.

246. Revell. 1970. 75.

247. Revell. 1970. 79.

248. For the interchange of [a] and [e] sounds see T-S A 19.11.

system.

Manuscript TS A 19.11 seems to represent early evidence for the usage of Tiberian vocalisation signs. However, it does not represent Standard Tiberian pronunciation. Thus, one could conclude that a stage of this tradition is transmitted where it has not been standardised.²⁴⁹

Moreover, the Palestinian does also give evidence of the pronunciation of *ševa mobile*. Although this notation is not coherent one can say that there are some tendencies. Usually it was reflected with an [e] sound and sometimes with an [a] sound. Whether the latter pronunciation shows Tiberian influence has still to be proven. Furthermore, the representation with an [i] sign shows Standard Tiberian pronunciation but reflects a ben Naftli vowel notation.²⁵⁰ The pronunciation with an [e] quality can be considered as typical Palestinian, since the Tiberian and most likely the Babylonian *ševa mobile* was realised as an [a] sound.

However, not only the Palestinian vocalisation shows some important features, but also the accentuation. The placing of accents not in accordance with their Tiberian counterpart is besides their graphical realisation the most striking observation. *Pašta*, for instance, is not always marked postpositive²⁵¹ like in the Tiberian system. As far as I can see, the Palestinian accentuation has not been placed according to the same rules as the Tiberian accentuation. Revell states that "the Palestinian accent signs do not originally mark stress position"²⁵² but "came to do so under the influence of the Tiberian system."²⁵³ Furthermore, he sees this influenced not imitated in a proper manner.²⁵⁴ Corresponding with my finds, Revell believes the irregular placement of Palestinian accents is showing possibilities, within which the accent may be placed. For this theory he is referring to Arabic, where, for example, in the word مملكة (*mamlaka* "kingdom")²⁵⁵ the first two syllables can be stressed but the third

249. Cf. Kahle (1959, 68), who found a manuscript with similar evidence in the JTS in New York.

250. Cf. explanations to T-S A 19.11.

251. Postpositive: The accent is written at the last letter of the word i.e. after the actual stressed syllable (e.g. *segholta*, *pašta*). Prepositive: The accent is written at the first letter of the word i.e. before the actual stressed syllable (e.g. *teliša*).

252. Revell. 1977. 201.

253. Revell. 1977. 201.

254. Revell. 1977. 201.

255. Cowan. 2011. 923.

cannot.²⁵⁶ Although, the same theory of stress placement cannot be applied to Palestinian Hebrew, Revell emphasizes that stress placement does not have to correspond to Standard Tiberian but might be, like in Arabic, applicable on more than one syllable of a word. This theory can also be shown on the manuscripts, that I have examined. The stress is in all cases placed at the ultima or the penultima, but never on the antepenultima or even before. Thus, I think, using the same argumentation, that the accents are placed on the stressed syllable. In T-S AS 63.59, a manuscript in shorthand-writing with mainly Tiberian vowel graphemes, *pašṭa*, for example, is also not marked postpositive but rather as described in the Palestinian system. This might show, that in shorthand-writing, the stress is placed on the initial letter of a syllable, that bears the stress in order to save space, since then only the first letter has to be written, whereas if it was placed postpositive also the last letter of the syllable had to be written. However, this is just an assumption, because there are exceptions in shorthand-writing where the whole syllable or word is written. Though, the placement of stress in such manuscripts shows, that the placement of the stress was fixed regarding the syllable but not regarding the place within this syllable. The same type of manuscripts with Babylonian vocalisation shows also the same phenomenon.²⁵⁷

As I have mentioned in the "Introduction" the accentuation system seems to be much older than the vocalisation system.²⁵⁸ For the origins of the Hebrew accentuation several options are taken into consideration. Revell²⁵⁹ and Merx²⁶⁰, for example, see some connection between the Hebrew accents and those in Syriac, whereas Praetorius believes that they are derived from Greek.²⁶¹ Dotan based on Arthur Spanier denies influence from these sources.²⁶² He assumes that the Hebrew accents already existed in the fourth or even the third century CE,

256. Revell. 1977. 203; On the Arabic stress placement see Fischer. 2006. 19-20 (§32).

257. Yeivin. 1962. 131.

258. Besides the references already mentioned in the "Introduction" see also Revell. 1977. 197.

259. Revell. 1977. 198.

260. Cf. Wickes, W. 1970 (Repr. 1881 and 1887). *Two Treatises on the Accentuation of the Old Testament* (with a Prolegomenon by Aron Dotan). Ktav: New York. xxxvi.

261. Wickes. 1970. xxxvi. Segal finds only partial influence of Greek accentuation on Syriac accentuation. He suggests, that the first accent, which appears in Syriac manuscripts (i.e. *pāšôqā*), is of Syriac origin and has parallels in other Semitic languages, whereas the recitation system is influenced by Greek (Segal. 2004. 63).

262. Wickes. 1970. xxxvii.

whereas the Syriac accentuation system had not been invented before the fourth century. Since Dotan is not giving any evidence for this assumption, it seems to as if he does not distinguish between the oral tradition, which is attested in the Talmud by the stories about the movement of the hands (see above), and the actual written signs itself, which is employed to fix the accentuation. However, in his later publication Dotan states that the accents "may well have been earlier than 500 CE",²⁶³ whereas Segal gives evidence of the development of accentuation signs in Syriac from around 400 CE.²⁶⁴ He identifies two accents at this time, which are regularly used and a few others, which are attested only seldom.²⁶⁵

The influence of Syriac vocalisation and accentuation on the Palestinian and the Hebrew vocalisation systems in general seems to me very likely. However, this thesis is only based on the usage of some signs that are used in the same or similar ways in both languages.²⁶⁶ Some Syriac scholars from the early Middle Ages connect the invention of Syriac accentuation with the translation of the Greek Bible into Syriac.²⁶⁷ This has to be treated with caution, since the Greek culture was regarded so high in general, that much of what was actually of Syriac origin was Graecized.²⁶⁸

Researchers like Morag suspect Syriac influence on the Hebrew vocalisation systems.²⁶⁹ However, it is difficult to find evidence in research papers or even books, that deal with this topic, since it appears to me that only little research has been conducted so far on in this field. Thus, I think it is worth to do further and more detailed research within this field.

Comparing Syriac with Hebrew regarding the vocalisation and accentuation traditions should be based on identical graphemes, which have been employed in the same way in both

263. Dotan. 1981. 98.

264. Segal. 2004. 58.

265. Segal. 2004. 58.

266. See above "Introduction" for examples. Cf. also Segal's explanations (2004, 58ff.). The smallest entity if the Syriac accentuation sign is the single dot. This is used in various combinations of maximum three dots to form further accents. One of them is š'wayyā, which is like the Tiberian *silluq* represented with the vertical dots at the end of a subdivision (Segal. 2004. 135). If one takes a closer look at the accentuation signs in Hebrew manuscripts like B19a, it becomes, graphically speaking, more difficult for the reader to identify the different accents, since the graphical distinction of accents suggested by Bible editions is not as clear as one might have wished.

267. Segal. 2004. 63; 66.

268. Segal. 2004. 63.

269. Morag. 1962. 46 (§3.2).

languages. It could be particularly interesting to investigate how the historical short [u] of Semitic languages developed in Syriac (and other Semitic languages) and the Palestinian system. Its reflection as [o] sound is in my opinion not based on the Standard Tiberian back [a] pronunciation (i.e. [ɔ]) of short *qamaṣ*.²⁷⁰ I think it rather reflects a more archaic pronunciation than Standard Tiberian. However, the notation of short [u] as [o] does not prove that the Palestinian pronunciation is older than the Standard Tiberian. There is no evidence apart from the increasing number of vowel signs that the Palestinian system is older than the Tiberian.

Furthermore, the thesis of the two groups of Palestinian Hebrew, which Revell has detected, could not only be verified by the evaluation of the codicological and palaeographical examination of the manuscripts but also by a detailed examination of the representation of *ševa mobile*.

270. Revell. 1970. 113.

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